Biomedical Knowledge Graphs and Large Language Models: A Critical Engineering Survey

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***Abstract*—Medical language models are fluent but hard to audit. Combined with language models, knowledge graphs offer structured, source-grounded facts. However, recent work often treats a knowledge graph as proof of grounding without verifying whether the system retrieved and used relevant subgraphs or produced clinically safe output. Our survey follows a small set of representative systems through their full lifecycle: graph construction and updating, indexing, retrieval, reasoning, verification, and post-deployment monitoring. We classify integration methods by where and how tightly graphs and models combine, distinguishing simple prompt-pasting from graph-aware encoders, and map the underlying ontologies, clinical datasets, and benchmarks supporting retrieval and tool-using agents.**

We find that graph benefits are highly conditional. Graphs help when they fill genuine knowledge gaps, entity linking succeeds, retrieval returns small relevant subgraphs, and evaluations use matched text-retrieval baselines. They help less and may hurt when coverage is thin, mappings fail, or retrieved context overpowers decisive evidence. We also examine practical failure modes, including weak citations, poor temporal handling, and closed-loop feedback, in which model proposals are written back into the graph and later retrieved as independent evidence.

***Index Terms*—Biomedical knowledge graph, large language model, graph retrieval-augmented generation, agentic systems, clinical artificial intelligence, evaluation, provenance.**

I. Introduction

BIOMEDICAL artificial intelligence works in a domain where knowledge is large, heterogeneous, contested, time-dependent, and consequential. A single clinical or scientific question can connect symptoms, diseases, drugs, genes, pathways, trials, guidelines, laboratory observations, and patient context. The relevant facts sit across terminologies, curated databases, publications, electronic health records (EHRs), and local protocols. They also differ in evidential strength, and they go out of date. These properties make biomedicine a natural but demanding setting for joining knowledge graphs (KGs) with large language models (LLMs).

A biomedical KG represents entities and typed relations in an explicit graph. Depending on its purpose it may encode ontology-backed concepts, literature-derived assertions, experimentally observed associations, patient events, or some mix of these. Resources such as Hetionet [1], SPOKE [2], the Clinical Knowledge Graph [3], PrimeKG [4], iKraph [5], and PubMed Knowledge Graph 2.0 [6] trace the field's move from individual relation datasets toward large, multimodal, provenance-bearing knowledge infrastructures. LLMs provide the complementary half: a general interface for interpreting questions, mapping language to concepts and tools, synthesizing evidence, and producing explanations. Their weaknesses matter just as much. Parametric knowledge can be stale or unverifiable, generated claims may be unsupported, confidence expressed in prose is poorly calibrated, and reasoning that looks plausible can hide a retrieval, logic, or data error.

The two components appear complementary. A graph can supply explicit semantics, neighborhood structure, paths, constraints, and source attribution. An LLM can translate natural language into graph operations, select and verbalize subgraphs, propose candidate relations, and lower the expertise needed to query a complex resource. This pairing has produced several method families: graph retrieval-augmented generation (GraphRAG), including deterministic Graph-RAG for mechanistically traceable drug-disease association discovery [7], natural-language-to-SPARQL or Cypher interfaces, graph-conditioned prediction via multimodal transformers [8], structured frequency-based graph traversal for drug effect classification [9], graph-aware fine-tuning, KG-constrained decoding, LLM-assisted entity and relation extraction, ontology alignment, graph completion, graph repair, and tool-using agents that alternate between retrieval, reasoning, and graph revision.

The intuitive case has too often stood in for empirical evidence. Retrieval can return an irrelevant path, and the model may ignore a relevant one. A fluent rationale may also be unfaithful to the path or the answer. These gaps have motivated deterministic Graph-RAG frameworks that prioritize mechanistic traceability in biomedical association discovery [7]. A citation likewise may not support the sentence it follows. Recent studies make these distinctions hard to ignore. A multi-model biomedical RAG study found the improvements small and inconsistent across datasets and retrieval settings, and it named evidence use, not retrieval alone, as the central bottleneck [10]. VERICITE found low sentence-level support among retrieved citation-claim pairs and showed that adding retrieval depth could make citation faithfulness worse [11]. MHGraphBench reported that a graph snippet can help one model and distract another, which makes graph textualization and context management part of the method rather than a neutral implementation detail [12]. Those three are peer-reviewed. A preprint temporal evaluation, ChronoMedKG, additionally reports a sharp fall from static to time-sensitive questions with

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graph retrieval recovering only part of the loss [13]; we treat its magnitudes as indicative and its direction as consistent with the peer-reviewed results above. These findings do not argue against graphs. They show that "KG-augmented" is an intervention whose mechanism and boundary conditions have to be measured.

The survey literature is itself getting crowded. Xu et al. [14] organize biomedical KG–LLM work around LLM-aided KG construction, KG-guided LLM enhancement, and applications. Murali et al. [15] separate KG-enhanced LLMs, LLM-augmented KGs, and synergistic systems. A systematic review and meta-analysis has already assessed retrieval-augmented generation across biomedicine [16], and preprint reviews cover biomedical RAG [17] and biomedical KGs more broadly [18]. Directionality is still useful. It is no longer a sufficient novelty claim or a sufficient analytical frame.

This survey therefore treats biomedical KG–LLM integration as an end-to-end engineered and governed lifecycle. Its contributions are fivefold.

1) *A multidimensional taxonomy.* We classify systems by knowledge-flow direction, lifecycle stage, coupling depth, knowledge object, and evidence maturity, rather than assuming that all "graph grounding" works the same way.
2) *An evidence-critical method synthesis.* We analyze graph-to-LLM, LLM-to-graph, and closed-loop systems in terms of mechanisms, dependencies, tradeoffs, failure propagation, and suitable baselines.
3) *A resource and benchmark map.* We compare biomedical KGs, ontologies, clinical data resources, tasks, and emerging graph-reasoning benchmarks, with explicit attention to licensing, provenance, versioning, temporal support, and target leakage.
4) *A layered evaluation framework.* We separate graph, retrieval, reasoning, generation, system, human, clinical, and deployment measures, and we specify the ablations and stress tests needed to establish that the graph itself causes a benefit.
5) *A translation and governance framework.* We introduce an evidence-maturity ladder and a hazard-control-verification model for provenance, privacy, security, interoperability, human oversight, change management, and circular self-validation.

The intended audience spans machine learning, knowledge representation, biomedical informatics, and clinical AI. KGs can improve biomedical LLM systems when their coverage, update policy, retrieval mechanism, representation, and verification process fit the task. They do not confer factuality, explainability, or safety by their presence alone. Fig. 1 summarizes the argument, and Fig. 2 maps the sections that develop it.

A. Relationship to prior surveys

Existing surveys organize the field mainly by the direction of knowledge flow, leaving several engineering and evidence questions unresolved. Table I compares the closest works and states what this survey adds.

This synthesis retains directionality but crosses it with lifecycle and coupling depth. Two systems can both be "KG-enhanced LLMs" while one merely pastes triples into a prompt and the other trains a graph-aware representation, runs iterative queries, verifies claims, and logs evidence versions. Grouping them without more resolution hides their engineering differences. The evaluation framework also locates where improvement or failure occurs. Evidence maturity and governance are treated as technical properties: version locking, provenance capture, rollback, access control, and workflow monitoring decide whether results can be reproduced and whether a deployed system can be controlled.

II. Review Scope and Method

The graph has to earn its place in a given system. Assessing whether it does requires clear definitions of a system, the evidence used to evaluate it, and the questions outside this review. This section defines those boundaries before the analysis begins.

A. Review type and objectives

This is a critical engineering survey, not a statistical meta-analysis. The literature mixes question answering, diagnosis, code prediction, link prediction, information extraction, drug repurposing, workflow support, and graph curation. Outcomes reported as exact match, F1, area under the curve, recall, clinician preference, or case-study plausibility are not commensurable, and pooling them would produce a precise-looking but invalid effect estimate. The goal instead is to identify architectures, define comparable engineering dimensions, weigh the strength of claimed benefits, and expose missing validation.

Foundational knowledge-graph and ontology work appears as background. The method synthesis concentrates on the period since instruction-tuned generative models became widely available, because that is when the coupling patterns discussed here took their current form.

B. Operational definitions

The inclusion boundary is as follows. A *knowledge graph* is an identifiable, explicit graph of entities and relations, normally with types, a schema or ontology, or graph-database semantics; a similarity network or a transient chain-of-thought diagram does not automatically qualify. An *LLM* is a pretrained generative or tool-using language or foundation model; encoder-only biomedical models appear only as direct technical precursors. A study must implement or evaluate a *functional interaction* between the graph and the language model, so mentioning both in the motivation is not enough. We distinguish curated biomedical KGs, literature-derived assertion graphs, ontology and terminology resources, EHR-derived patient graphs, and transient task-specific graphs, because their provenance, privacy, and validation requirements differ. We also separate scientific discovery, information access, clinical research operations, clinician-facing support, and patient-facing communication, because success in one stratum is not evidence of readiness in another.

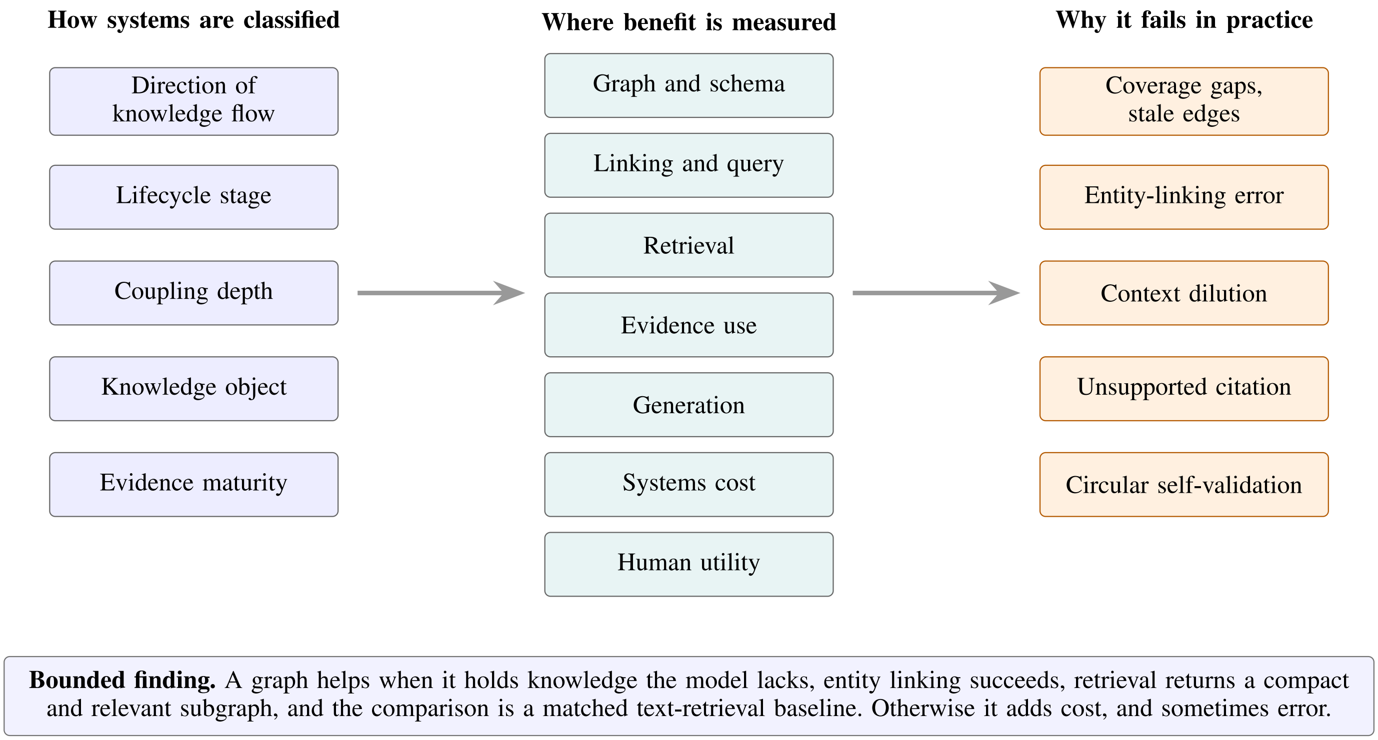

Fig. 1. The structure of this survey's argument. **Left:** systems are classified along five axes rather than by direction of knowledge flow alone, because pasting triples into a prompt and training a graph-aware encoder are different interventions that the usual taxonomy treats alike. Coupling depth runs from C0, where graph and model share an application but exchange nothing structured, to C5, where the graph constrains decoding and may receive controlled write-back; evidence maturity runs from E0, a prototype with no systematic evaluation, to E6, governed routine use. **Centre:** benefit is measured at seven separable layers, L1 through L7 in the order shown, so that a failure can be attributed to the graph, the linker, the retriever, or the generator rather than to the system as a whole. **Right:** the failure modes that recur across the reviewed systems. Circular self-validation arises only in closed-loop designs that write model output back into the graph, and it has no analogue in unidirectional systems.

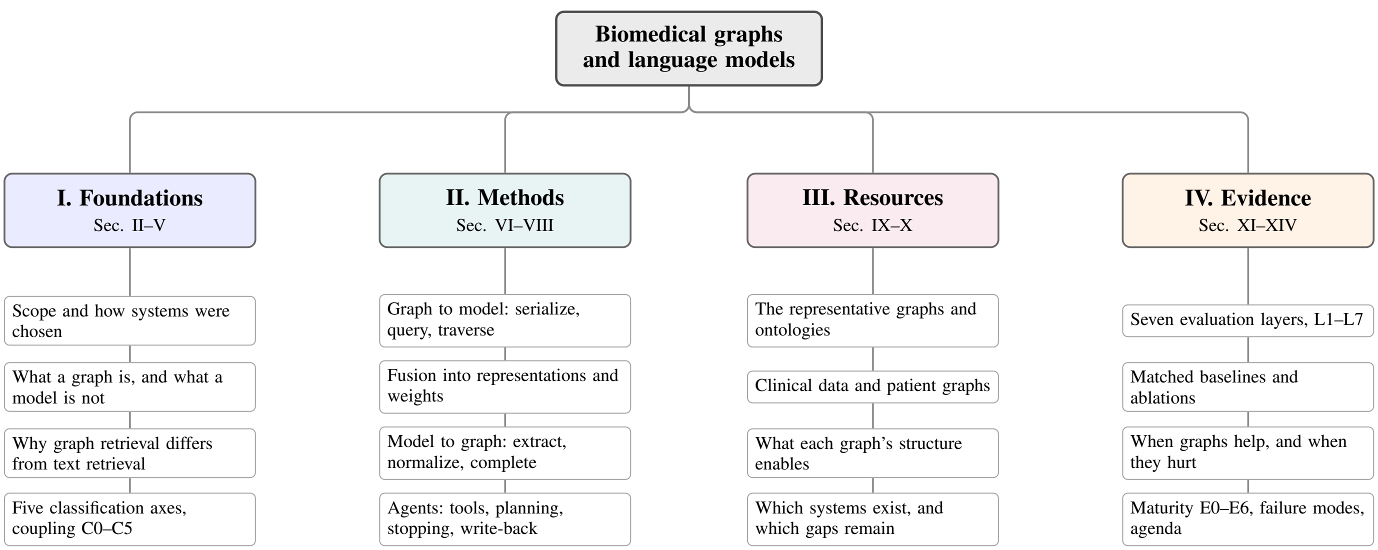

Fig. 2. **How the survey is organized, and what to read for a given purpose.** Part I settles definitions and introduces the two graded scales the rest of the paper uses, coupling depth and evidence maturity. Part II covers the methods themselves in both directions, from graph to model and from model to graph, ending with agents that do both. Part III is the reference material: which graphs exist, what each one's structure makes possible, and which systems have been built on each. Part IV is the critical apparatus, covering how to measure a benefit, when the benefit appears, how mature the clinical evidence is, and how these systems fail. A reader choosing a graph for a new system should go to Part III first, then Part II for the coupling that fits the task. A reader reviewing someone else's claim should start at Part IV.

C. How systems were selected

This survey is built around representative systems rather than exhaustive coverage. An engineer designing a graph-grounded clinical system should be able to find the two or three papers closest to the intended setting and understand what they did, what they proved, and what they left open. A

TABLE I
HOW THIS SURVEY DIFFERS FROM THE CLOSEST EXISTING SYNTHESES. PRIOR REVIEWS ORGANIZE THE FIELD MAINLY BY THE DIRECTION KNOWLEDGE FLOWS, THAT IS, WHETHER THE GRAPH IMPROVES THE MODEL OR THE MODEL IMPROVES THE GRAPH. THAT DISTINCTION IS REAL BUT COARSE: IT GROUPS A SYSTEM THAT PASTES TRIPLES INTO A PROMPT WITH ONE THAT TRAINS A GRAPH-AWARE ENCODER, VERIFIES CLAIMS, AND VERSIONS ITS EVIDENCE. THE RIGHTMOST COLUMN IDENTIFIES THE ENGINEERING QUESTIONS THAT REMAIN AFTER DIRECTIONALITY IS ACCOUNTED FOR.

Synthesis	Principal scope	What it establishes	Remaining gap addressed here
Xu et al., KDD 2025 [14]	Biomedical LLM–KG unification	Construction, KG-guided enhancement, applications	Lifecycle failure propagation, evaluation layers, maturity, governance
Murali et al., IEEE JBHI 2026 [15]	Medical KG–LLM integration	KG-enhanced, LLM-augmented, synergistic categories	Coupling depth, conditional benefit, validity threats, deployment controls
Liu et al., JAMIA 2025 [16]	Biomedical RAG review and meta-analysis	Evidence on retrieval augmentation and clinical development	Graph-specific retrieval, construction, representation, constraints, updates
He et al., 2025 [17]	Biomedical RAG pipeline	Retrieval, reranking, generation, datasets, applications	KG lifecycle and graph-specific failure modes beyond RAG
Lu et al., 2025 [18]	Biomedical KG landscape	Domains, tasks, resources, applications	Functional coupling to generative models and comparative controls
GraphRAG survey, 2025 [19]	General graph RAG	Broad technical taxonomy across domains	Biomedical semantics, evidence, clinical risk, translational maturity

list of ninety systems with one sentence each conveys less than a dozen analyzed properly.

A system earned a place when it did at least one of the following: introduced a coupling mechanism that others have since built on; exposed a failure mode that changes how the class of systems should be evaluated; established a resource or benchmark that later work depends on; or carried its evaluation past offline benchmarking into clinician-facing or prospective use. Where several papers make the same architectural move, we discuss the clearest instance and treat the rest as variants. Papers were left out when the "graph" was a document similarity network with no typed semantics, when the model was neither generative nor tool-using, when the biomedical setting was incidental, or when the graph-model interaction was described but not implemented.

The sampling frame behind those judgments came from five conjunctive queries run against arXiv, PubMed, OpenAlex, and DBLP in August 2026. The searches returned 1879 records and 1147 distinct works after deduplication; 105 of those works are cited here. Screening was purposive rather than systematic, and the authors applied the criteria above without independent duplicate assessment. We therefore report the frame and criteria rather than a stage-by-stage exclusion count. The full query strings, per-interface yields, and the verification procedure applied to every retained reference are given in the supplementary material [20].

Peer-reviewed work carries the argument. Preprints appear where they introduce a resource or a result that changes what the field currently believes, and they are labeled as preprints at the point of use so that a reader can weigh them accordingly.

D. What this survey does not cover

Three boundaries are worth stating. The review does not attempt a quantitative meta-analysis because the outcomes are not commensurable, as discussed above. Nor does it rank systems by headline score, since those comparisons are usually confounded by corpus, generator, and context budget. Vendor-specific model performance is treated as tied to a particular model version rather than as a durable property of the architecture, since a hosted endpoint can change without notice.

E. Evidence extraction and appraisal

For each representative system the synthesis records the KG name and version, graph source and provenance, schema, temporal properties, model and version, coupling mechanism, graph-query and retrieval stages, task and intended user, dataset and split, comparator, outcome, ablation, evidence status, code and data availability, and any operational or clinical validation. Appraisal is a profile, not a single quality score. A study can have a rigorous retrieval evaluation and weak clinical generalizability, or strong predictive discrimination and poor leakage control.

Four validity threats get special attention. *Graph-to-label leakage* occurs when labels, answers, or answer-derived paths sit in the inference graph. *Temporal leakage* occurs when a future graph snapshot answers a historical question. *Model contamination* occurs when a benchmark or its solutions may have appeared in pretraining. *Evaluator circularity* occurs when an LLM judges the outputs of a related LLM without independent validation. These threats matter more than minor prompt differences and should be documented.

III. FOUNDATIONS

The rest of this survey turns on properties that a triple store does not automatically have: qualifiers, provenance, time, and a release process. This section establishes those properties for graphs and for language models separately, because the failure modes analysed later are almost always a mismatch between the two rather than a defect in either.

A. Biomedical knowledge graphs as evidence-bearing infrastructure

A KG can be written as a set of triples (h, r, t), where a head entity h and a tail entity t are joined by a relation r. Biomedical graphs usually need more than this bare form. An assertion may require qualifiers such as population, dose, tissue, experimental condition, publication, confidence, valid time, transaction time, or negation. An edge such as *drug treats disease* means something different if it denotes an approved indication, a trial hypothesis, an observational association, a computational prediction, or a statement pulled from a retracted article. Reification, named graphs, hyperedges, statement identifiers, or property graphs can keep these distinctions. Flattening them into unqualified triples creates false equivalence.

Schemas and ontologies constrain meaning. UMLS integrates many biomedical vocabularies [21]; SNOMED CT supports clinical concepts and relations [22]; RxNorm normalizes drugs [23]; LOINC identifies observations [24]; the Human Phenotype Ontology represents phenotypic abnormalities [25]; MONDO integrates disease concepts [26]; the Gene Ontology describes gene-product functions [27]; and ChEBI represents chemical entities [28]. Their role extends beyond terminology. They fix entity granularity, allowed relations, subsumption, logical constraints, and interoperability, and they carry license and update obligations that flow downstream.

Biomedical graphs differ along six dimensions, and each one changes what a system built on them can do. *Construction*: expert curated, database integrated, literature extracted, EHR derived, or model generated. *Semantics*: ontology-backed, schema-light, property graph, RDF, or a task-specific heterogeneous network. *Evidence*: provenance at the source level, at the assertion level, as a confidence code, or not at all. *Time*: a static release, periodic versions, valid-time aware, longitudinal, or continuously updated. *Scope*: general, molecular, pharmacological, disease-specific, or patient-specific. *Access*: open, registration controlled, institution restricted, or licensed.

These are causal inputs to system behavior, not catalogue entries. A literature-extracted graph offers scale and inherits extraction error and publication bias. A curated graph offers stronger semantics and lags new evidence. An EHR graph carries local clinical context along with coding artifacts, missingness, local practice patterns, and protected information. Whatever the model retrieves, it is grounded only in what that graph happens to represent.

B. Large language models as probabilistic interfaces and controllers

LLMs generate tokens conditional on context. Instruction tuning, preference optimization, long-context mechanisms, retrieval augmentation, structured decoding, and tool use have turned them from text generators into general interfaces and workflow controllers. In biomedical systems they may perform concept normalization, query planning, evidence selection, synthesis, explanation, and conversational interaction. Domain adaptation can improve terminology and task behavior, but it does not establish access to current, sourceable evidence.

Domain-pretrained generative models complement a graph rather than replace it. Med-PaLM demonstrated expert-level performance on medical examination benchmarks from parameters alone [29]. Open models such as MEDITRON-70B [30] and BioMistral [31] narrowed that gap with published weights. Their local deployment also permits version pinning in a way that a hosted endpoint may not. Domain pretraining improves terminology, calibration on in-domain phrasing, and instruction following, but the knowledge remains undated and unattributable, and updating it requires retraining. A stronger biomedical backbone raises the closed-book baseline that any graph-augmented system must beat, making the matched comparison of Section X more demanding. The reviewed evidence that graph gains shrink as backbones strengthen [32] is consistent with this effect.

Four kinds of knowledge coexist in such systems. *Parametric knowledge* is encoded in weights and is hard to inspect or update selectively. *Contextual knowledge* is the documents, triples, paths, or tables placed in the prompt. *Tool-accessed knowledge* is returned by graph queries, search systems, calculators, or clinical services during execution. *Committed knowledge* is what gets written into a graph, record, report, or decision-support workflow. Risk climbs as information moves from parametric or contextual use into committed knowledge. A hallucinated relation in an exploratory answer is a quality failure. The same relation written to a production KG and later retrieved as established evidence is a persistent systems hazard.

C. Why graph retrieval is not simply text retrieval

Conventional text RAG retrieves passages by lexical or embedding similarity. Graph retrieval instead exploits typed entities and relations, neighborhoods, paths, communities, schema constraints, or combinations of graph and text indexes. The unit of evidence may be a triple, an evidence-bearing edge, a path, a bounded subgraph, a community summary, or a graph-linked passage. This supports compositional questions and explicit relational constraints, but it adds failure points:

$$\begin{aligned} P(\text{correct answer}) &\leq P(\text{correct linking}) \cdot P(\text{coverage}) \\ &\quad \cdot P(\text{retrieval} \mid \text{linking}) \cdot P(\text{faithful use} \mid \text{evidence}). \end{aligned} \tag{1}$$

The expression is illustrative rather than an independence assumption. It shows why improving only the generator cannot overcome missing coverage, and why a high-quality graph cannot rescue poor entity linking or poor evidence use. Graph traversal can recover multi-hop relations a passage retriever misses, yet unconstrained expansion produces noisy neighborhoods and eats context. Shortest paths are not automatically the most evidentially relevant. Central nodes can dominate community methods. Textualizing triples can strip qualifiers, and graph embeddings can obscure provenance. The graph interface is therefore a first-class component to evaluate.

D. Grounding is not the same as verification

Four properties get run together under the word "grounded," and separating them is what makes the evaluation framework of Section X possible. An output may be conditioned on an external source, and a claim may be linked to a retrievable assertion. Separate questions are whether the output follows from that assertion and whether the information helps a user decide or verify. A visible graph path provides attribution without establishing entailment or user utility, and a fluent rationale can be assembled after the answer was already chosen. Verification can be symbolic, retrieval-based, statistical, model-based, human, or some combination, and a biomedical system should say which property it verified and which it left alone.

E. The end-to-end lifecycle

The unit of analysis in this survey is the whole system in Fig. 3, not the model in the middle of it. Every transition can fail, and the failures travel. A construction error becomes a retrieval candidate. A query error selects the wrong subgraph. A rendering error erases the qualifier that mattered. Weak verification passes an unsupported claim. Unsafe feedback turns today's output error into tomorrow's ground truth. That is why a trustworthy design needs observability and version control along the whole chain, not just at the model.

IV. A Multidimensional Engineering Taxonomy

Sections V through VII analyse a large number of systems, and a directional label alone does not distinguish them. The five axes below give the vocabulary those sections use. They are engineering constructs proposed to organise evidence and support comparison, not validated measurement instruments. The levels are ordered and each has a boundary test, but they carry no psychometric claim; a system's position describes its architecture and evidence rather than assigning a score. Where a system straddles two levels, the convention throughout is to assign the deeper level only if the mechanism is load-bearing for the reported result.

A. Axis 1: direction of knowledge flow

The familiar directional split is still a good entry point. In *KG→LLM* systems the graph supplies knowledge, structure, constraints, training signals, or verification to an LLM. In *LLM→KG* systems the model extracts, normalizes, completes, aligns, repairs, summarizes, or updates a graph. In *bidirectional* systems the two alternate, sometimes writing reviewed outputs back to the graph. Direction describes what flows, not how, so it should never be the only label.

B. Axis 2: lifecycle stage

Systems may couple at source acquisition, graph construction, normalization, storage, indexing, query interpretation, retrieval, reasoning, generation, verification, user interaction, or monitoring. The stage changes the claim. An LLM used to normalize synonyms can improve graph coverage but does not directly ground a downstream answer. A KG consulted only after generation can catch contradictions but does not guide the initial reasoning. A system is best described by a lifecycle signature rather than a single tag.

C. Axis 3: coupling depth

We distinguish six coupling levels, shown in Fig. 4. *C0, co-presence*: the graph and LLM share an application but exchange no structured information; this is not substantive integration. *C1, prompt or interface coupling*: triples, paths, summaries, or query results enter a prompt, or the LLM emits a graph query; flexible and auditable, but sensitive to rendering and context limits. *C2, iterative tool coupling*: the LLM plans, calls graph tools, observes results, and revises; more adaptive, but with trajectory and agent-control risk. *C3, representation coupling*: graph embeddings, GNN outputs, graph tokens, or adapters fuse with language representations; compact, but harder to attribute and update. *C4, objective or parameter coupling*: graph-derived examples, paths, constraints, or contrastive objectives change the weights; can improve internal behavior, but weakens source-level traceability and selective updating. *C5, constrained or closed-loop coupling*: the graph constrains decoding or verifies output, and outputs may trigger controlled graph revision; stronger control only when the verifier is independent and updates are quarantined.

Five questions separate the levels, and they can be answered from a system description without judgment. Does structured graph content reach the model at inference time, or a graph query leave it? If not, the system is C0. Does the model issue more than one graph call within a single answer, conditioned on what earlier calls returned? If so, C2 rather than C1. Are graph-derived vectors, tokens, or adapter modules combined with language representations? That is C3. Were model weights changed using graph-derived supervision, examples, or constraints? That is C4, and it is the only level that cannot be reversed by editing the graph. Does the graph gate what may be emitted, or does output feed back into the graph? That is C5. A system can satisfy several tests at once, in which case the convention above applies and the deeper level is recorded only when it carries the reported result.

Coupling depth is not a quality ranking. C1 may be preferable when auditability and fast knowledge updates matter. C3 and C4 may win on latency or high-volume prediction. C5 can be the safest or the most dangerous option depending on its update controls. The choice is driven by what the task must expose. C1 and C2 keep evidence inspectable and the graph correctable when a clinician must audit each claim or the underlying knowledge changes faster than a training cycle. C3 and C4 are defensible when throughput or latency dominates and attribution is not required at the point of use, as in high-volume triage or molecular property prediction. An output that must satisfy a written rule, such as an eligibility or interaction constraint, favors C5 with a deterministic verifier over generator-only enforcement. Section IX works this through for particular graphs and tasks, and Table X records the coupling depth appropriate to each application area.

D. Axis 4: knowledge object

The graph object may be an ontology or terminology graph, a curated factual or evidence graph, a literature-derived assertion graph, a molecular or drug–disease graph, a patient-specific longitudinal graph, a multimodal graph linking

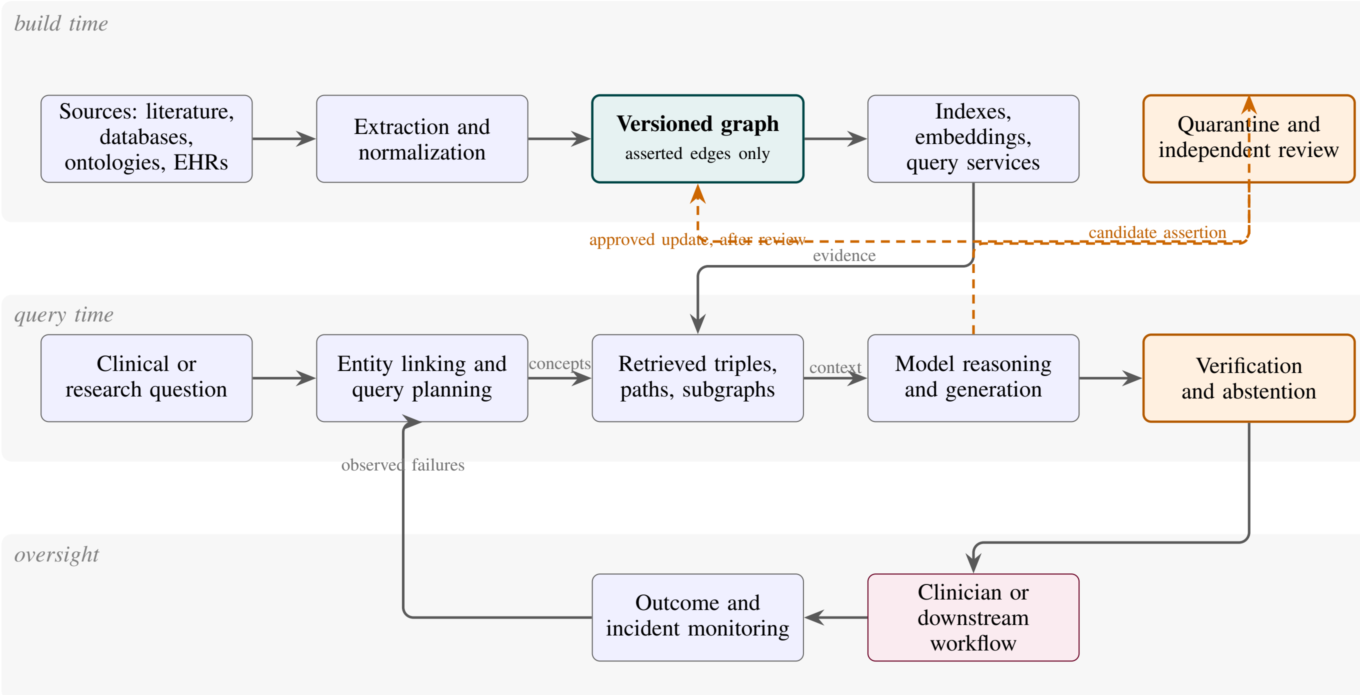

Fig. 3. **The path a graph-grounded answer travels, and the single point at which model output is allowed back into the graph.** The bands separate work done before any question is asked, work done per query, and the oversight that closes the loop. Solid arrows carry evidence forward. Sources become normalized edges, edges become a versioned graph, the graph is indexed, and retrieval hands a subgraph to the model. Two return paths matter. Monitoring feeds observed failures back into linking and planning, which is how a system improves without its knowledge changing. The dashed orange path is the only route from model output to the graph. A proposed assertion is quarantined, independently reviewed, and only then committed. Without that detour, the model's own guess returns later as apparently independent evidence, creating the circular self-validation failure of Section XII. The graph is labelled "asserted edges only" to mark the boundary that keeps unreviewed candidates out of retrieval.

text, images, omics, and clinical events, a task-local graph built from retrieved documents, or a community or summary graph derived from a larger resource. This axis explains why a method validated on an open molecular graph cannot transfer uncritically to a private patient graph. The former emphasizes scientific evidence and link-prediction validity; the latter adds privacy, missingness, coding drift, and workflow consequences.

E. Axis 5: evidence maturity and clinical risk

We classify evidence from E0 (conceptual) through E6 (routine monitored deployment), defined in Section XI-G. Clinical risk is graded separately by intended use: research support, administrative or informational support, clinician-facing recommendation, or patient-facing decision influence. A high-performing E1 prototype aimed at a high-risk use is still immature.

F. Taxonomy in practice

Table II expresses common method families along these axes, together with their primary benefit and their principal failure mode. The rest of the survey uses this taxonomy to compare methods while keeping the distinctions that decide reliability.

V. KG-to-LLM Methods: From Retrieved Facts to Controlled Generation

The KG-to-LLM direction is the most visible part of the literature, and the label hides several different things. The graph may be turned into text, traversed as a symbolic tool, compressed into learned representations, used to supervise training, or consulted after the answer is written. These fix different bottlenecks. Pooling them as one method is how the field ends up arguing about whether "KG-RAG works." Fig. 5 reduces four of the recurring designs to their essential blocks.

A. Serializing the graph into the prompt

The simplest interface retrieves a subgraph and drops a text rendering of it into the prompt. The appeal is modularity. The graph store and the model can be swapped independently, and whatever was retrieved can be logged. The difficulty is organization. One-hop expansion grows fast around a high-degree concept such as *neoplasm*, and an unordered list of triples hides which relations form a path.

SPOKE-based KG-RAG prunes hard, using prompt-aware extraction and embedding-based selection to cut token use by more than half with no loss on the tested tasks [33]. That is efficient evidence packaging, not proof that the model reasons symbolically. MindMap asks the model to emit a graph-like structure before it answers, which helps inspection [34]. But an emitted path is still model output. It becomes a faithful trace only when each edge links back to something retrieved. ILlama

TABLE II

The common method families, placed on all five classification axes. Each row is a way of joining a graph to a language model. *Direction* says which component supplies knowledge to the other. *Lifecycle stage* says where in the pipeline the exchange happens, since a model used to normalize synonyms during construction makes a different claim from one consulted at query time. *Coupling* gives the depth level defined in Fig. 4. The final columns state the primary benefit and characteristic failure. The families differ in how they fail, so pooling them under one label hides important distinctions.

Family	**Direction**	**Lifecycle stage**	**Coupling**	**Primary benefit**	**Principal failure**
Triple/path prompting	KG→LLM	Retrieval/generation	C1	Explicit, inspectable evidence	Context noise; qualifier loss
Natural-language graph query	Bidirectional	Query/retrieval	C1–C2	Accessible structured search	Invalid or semantically wrong queries
Community GraphRAG	KG→LLM	Index/retrieval	C1–C2	Broader multi-hop context	Hub bias; lossy summaries
Graph encoder or adapter	KG→LLM	Representation/inference	C3	Compact structural signal	Weak attribution; update coupling
KG-derived instruction tuning	KG→LLM	Training	C4	Internalized domain behavior	Staleness; source opacity
Graph-constrained decoding	KG→LLM	Generation/verification	C5	Schema or relation validity	Incomplete graph suppresses valid novelty
LLM extraction and normalization	LLM→KG	Construction	C1–C4	Scale and schema mapping	Hallucinated or duplicated assertions
LLM graph completion	LLM→KG	Curation/update	C1–C4	Candidate hypothesis generation	Plausibility mistaken for evidence
Iterative graph agent	Bidirectional	Query/reasoning	C2–C5	Adaptive multi-step search	Tool errors and trajectory instability
Self-updating graph system	Bidirectional	Monitoring/update	C5	Continuous learning	Circular self-validation and poisoning

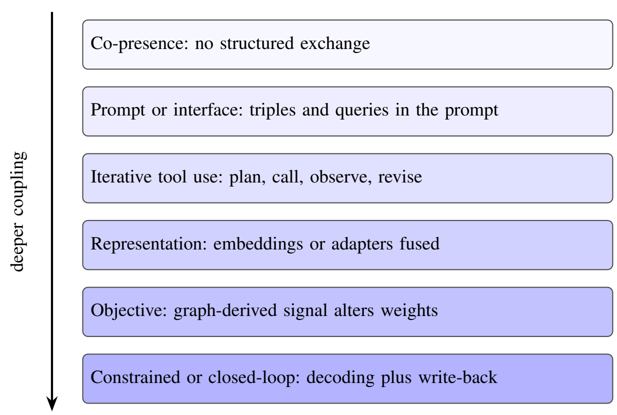

Fig. 4. **Coupling depth: how tightly a knowledge graph and a language model are joined.** Depth describes where in the system the two components exchange information, not how much knowledge flows. At C0 they share an application but pass nothing structured to each other, which we include only to mark the boundary of genuine integration. At C1 the graph reaches the model as text in the prompt, or the model emits a query; evidence stays inspectable, and results depend heavily on how triples are rendered and on the context budget. At C2 the model calls graph tools in a loop, observing results and revising, which adds adaptivity and the risk of runaway trajectories. At C3 graph embeddings or adapters are fused into the model's representations, and at C4 graph-derived signal changes the weights themselves. Both improve speed and compactness while reducing source-level attribution, because a projected vector is not a citation. At C5 the graph constrains decoding and may receive reviewed write-back, which is the tightest control available when the verifier is independent and updates are quarantined, and the largest hazard when they are not. Depth is not a quality ranking. Shallow coupling is preferable when a clinician must audit each claim; deep coupling is preferable when latency or throughput dominates.

precomputes UMLS triple documents and retrieves over those, which moves graph expansion off the latency-critical path [35]. The cost is rigidity: change the graph and the derived corpus has to be rebuilt.

The shared lesson concerns what survives compression. Serialization suits small subgraphs where provenance must stay visible and the model may be replaced. It fails when topology carries the task signal, when many parallel paths must be compared, or when qualifiers cannot survive being turned into prose, because a verbalized triple can quietly upgrade an association into a cause. A fair evaluation compares at least three organizers: unordered triples, explicit paths, and source passages, under one token budget.

B. *Letting the model write the query*

Instead of shipping a neighborhood to the model, a system can have the model write a formal query. This provides exact filters, relation constraints, and joins while keeping the graph current without retraining. Federated bioinformatics work retrieves schemas and example queries to generate SPARQL, validates the queries, and attempts repair on failure [36]. BioChatter exposes graph schemas to conversational models over BioCypher graphs [37].

Both illustrate the same limit: execution feedback catches a malformed query and cannot catch a valid query that encodes the wrong intent. Query generation therefore needs three separate checks. Syntactic executability and schema validity come free from the database. Semantic fidelity to the question does not, and testing it requires gold query intent alongside adversarial synonyms, negation, temporal constraints, and ambiguous concepts. For patient data there is a fourth requirement that is easy to miss: access control has to run before execution,

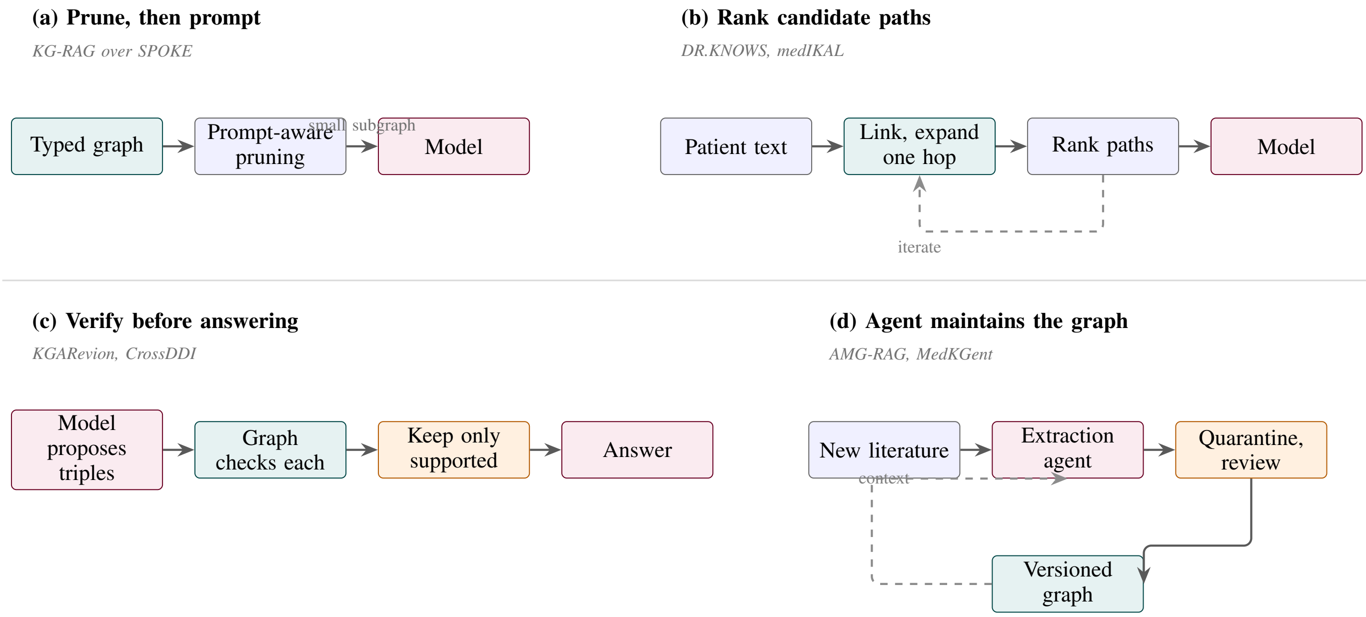

Fig. 5. **Four recurring architectures, reduced to their essential blocks.** Colour marks the role each component plays: teal for the graph, purple for the language model, and orange for a control that can reject a candidate. (a) The graph is pruned before it reaches the prompt, so the model sees a small subgraph rather than a neighbourhood. Pruning determines which evidence remains visible. (b) Patient text is linked to concepts, the graph is expanded a hop at a time, and candidate paths are ranked before generation. The dashed arrow marks the iteration that makes seed-concept errors so costly. (c) The order is inverted: the model proposes candidate triples and the graph acts as a filter, so the model contributes recall and the graph retains authority. Its ceiling is graph completeness, since a true but absent claim is rejected. (d) The agent's job is the graph itself. The orange review block prevents an uncontrolled loop by requiring an extracted claim to survive review before it becomes retrievable evidence.

because a correct query can still return records the user may not see.

C. Path and subgraph retrieval

Path methods exploit entity types, clinical context, and path semantics, and the four most instructive systems fail in four different ways.

KRAGEN decomposes a question through graph-of-thought prompting and retrieves per subproblem [38]. The decomposition helps but commits the method early: if the first plan misses the decisive mechanism, later retrieval cannot recover it. Plan diversity and recovery from empty retrieval are therefore more informative than final accuracy alone. DR.KNOWS ranks candidate UMLS paths with a graph isomorphism network and iterates toward diagnosis concepts, reporting gains on MIMIC-III and an institutional dataset. Its error analysis identifies incorrect starting concepts as a recurring cause, providing clear evidence that entity linking belongs inside the model rather than in preprocessing [39]. medIKAL merges an initial model diagnosis with graph search and reranks paths [40], recovering diseases absent from the retrieved paths while blurring attribution, since an answer may come from the graph, the initial guess, or their interaction. HyKGE inverts the order, generating a hypothesis first and using it to steer exploration [41], which handles underspecified questions and invites confirmation bias, because a wrong hypothesis steers retrieval toward paths that support it.

At the molecular scale, KGRxn-LLM does something the clinical papers should copy: its benchmark separates questions whose answers the graph covers from held-out questions requiring generalization beyond it [42]. Reporting one aggregate score over both conflates retrieval competence with generalization.

Across all of these, retrieval has three independent ceilings: correct seed concepts, adequate coverage, and an effective search policy. Adding hops addresses only the search policy and usually worsens noise and cost. A comparison that does not report coverage conditional on each factor, with oracle-seed and oracle-path upper bounds, cannot identify the limiting stage.

D. Community, hierarchy, and hybrid retrieval

Path retrieval answers local questions and struggles with broad ones. Community methods, following the pattern set for query-focused summarization [43], give the retriever something to return when no single path decides the answer.

KARE detects hierarchical communities over a multi-source concept graph, retrieves patient-relevant ones, and distills reasoning traces into a smaller model, reporting gains on mortality and readmission prediction [44]. Because it couples community indexing, patient-context augmentation, trace distillation, and prediction, the gain cannot be attributed to topology without ablating each component. Its label-derived traces also need a leakage check, since the expert model sees ground-truth outcomes during training-sample construction. MedRAG takes the opposite approach and shapes the hierarchy to the task, encoding discrimination among diseases with overlapping presentations rather than generic similarity [45]; the design is clinically apt and correspondingly narrow, and its private evaluation limits replication. MedGraphRAG links user documents to controlled terminology and authoritative sources, combining precise and broad retrieval [46], which answers a real weakness of pure graph retrieval: edges identify relationships while passages retain the qualifiers. HippoRAG 2 formalizes this dual granularity by coupling phrase-level indexing with Personalized PageRank across open-vocabulary entity graphs [47], enabling multi-hop associative recall without predefined community clustering; however, its reliance on dense phrase extraction and graph traversal makes it sensitive to graph density and noisy edge formation over long clinical narratives.

Together they imply a routing principle rather than a winner. Paths suit relation-specific and mechanistic questions, associative walks and communities suit synthesis and patient-context enrichment, and hybrid graph-and-text packets suit anything where qualifiers or citation entailment matter. One retriever need not serve every question, though a router's own errors then need measuring.

General-purpose GraphRAG engineering has meanwhile moved faster than its biomedical application. LightRAG replaces global community summarisation with a dual-level keyword index and incremental insertion, cutting indexing cost and allowing a graph to be updated without a full rebuild [48]. HippoRAG uses a schema-less open information extraction graph with personalised PageRank to perform associative multi-hop retrieval in a single step [49]. Both target the cost and staleness problems that matter most in biomedicine, where the graph changes underneath the index, and neither has been evaluated on a curated biomedical graph with typed relations. The transfer is not automatic: LightRAG's incremental index assumes edges are additive rather than retracted, which a terminology release violates, and HippoRAG's extracted graph has no ontology grounding and therefore none of the qualifier or provenance structure that Section IX shows to be the source of biomedical value. Adapting these efficiency gains to versioned, typed, provenance-bearing graphs is an open and tractable problem.

E. Fusing graph structure into the model

Textual retrieval keeps evidence inspectable and spends context. Representation-level methods project structure into the model's latent space instead.

BioBridge learns graph-supervised transformations between independently trained foundation models without fine-tuning the encoders, supporting cross-modal retrieval [50], which establishes that a graph can supervise mappings between modalities rather than merely store text. MEG projects graph embeddings into an instruction-tuned model through a learned projection, gaining on four medical multiple-choice datasets [51]. MedCo runs the coupling in both directions, building a medical-code graph from EHR associations and type-constrained inferred relations, then jointly training a LoRA-tuned encoder with a heterogeneous GNN [52]. MedKInstruct moves the graph from inference-time context to training-time supervision, using graph paths to synthesize visual question-answer pairs and reward path consistency [53].

The common cost is attribution. A projected vector is not a citation, and a model can exploit correlations encoded in an embedding without exposing the supporting path. MedCo's generated rationales are training signals rather than evidence, and MedKInstruct's instruction synthesis opens a leakage route, since benchmark answers present in the graph can be reproduced without attending to the image. Deep coupling earns its place when latency or context rules out explicit evidence, or when the task needs cross-modal alignment. It does not replace retrieval when someone has to name the source behind a clinical claim.

F. Verifying after generation

The last family separates fluent generation from acceptance, and it contains the strongest safety argument in this literature.

KGARevion has the model propose query-relevant triples, then checks them against a grounded graph and generates only from what survives [54]. Parametric knowledge raises recall while the graph keeps authority, and the ceiling is graph completeness, since a true but absent candidate is rejected. "Verified" therefore means supported by this snapshot, not true. CrossDDI goes further and removes the model from arbitration entirely, issuing a positive drug-interaction verdict only on explicit supporting evidence, which reduced variation across three backbones while exposing evidence acquisition as the binding constraint [55].

The distinction that matters is between generative explanation, graph consistency, and evidence entailment. A response can satisfy graph consistency and cite nothing, and a cited source can fail to entail the sentence attached to it. VERICITE found sentence-level support for only 27 to 41% of citation pairs at common retrieval depths, and supplying gold documents improved answers more than citations, which locates the problem in generation rather than retrieval [11]. A system reporting a single "verified" flag is not measuring any of the three separately.

VI. LLM-to-KG Methods: Construction, Alignment, Enrichment, and Completion

LLMs lower the cost of graph creation by replacing task-specific extraction models with schema-guided prompting and by generating descriptions or candidate relations. Their fluent output also makes malformed or unsupported structure easy to miss. The engineering challenge is to normalize, qualify, validate, version, and promote candidate triples safely.

A. Schema-constrained extraction and semantic normalization

SPIRES is an early and influential pattern. A user supplies a LinkML schema, the system recursively prompts an LLM to fill nested fields, and it then grounds extracted values to existing ontologies through external normalization tools. It supports diverse biomedical schemas without task-specific training and is implemented in OntoGPT [56]. The separation between generative extraction and ontology grounding is the key idea, because it accepts that an LLM may recognize a relation linguistically while failing to assign a stable identifier.

Arsenyan et al. [57] tested twelve language-model architectures on extracting diseases, factors, treatments, and manifestations from EMR notes, scoring both structured output and hallucination against manual annotations. Bigger decoder-only models did not dominate. Extraction is still a constrained prediction problem, and smaller supervised models can be competitive, cheaper, and easier to validate.

RELATE handles what happens after free-form relation extraction. It embeds an ontology's predicates, retrieves candidate Biolink relations with SapBERT, and uses an LLM to rerank them while handling negation. On ChemProt it reported 52% exact match but 94% accuracy at rank ten, which shows candidate retrieval was much stronger than final relation selection [58]. That gap is diagnostically useful, because it separates ontology coverage and embedding retrieval from LLM disambiguation. Construction studies should report entity detection, identifier grounding, predicate candidate recall, final predicate accuracy, qualifier extraction, and negation as separate stages.

Schema constraints solve structural validity, not factual validity. A perfectly valid triple can reverse subject and object, erase negation, omit species, or turn correlation into causation. Promotion into an asserted biomedical KG should therefore require evidence spans, source identifiers, confidence calibration, and relation-specific validation rules.

B. Human-in-the-loop construction at scale

The Mental Disorders Knowledge Graph (MDKG) is the strongest recent example of an LLM embedded in a larger curation pipeline rather than used as an autonomous extractor. The workflow combines GPT-4-assisted pre-labeling, active-learning annotation, named-entity recognition, relation extraction, database parsing, and LLM-based triple refinement. Its relations retain conditional statements, cohort characteristics, provenance, and confidence, and the published graph contains more than ten million relations, including nearly one million associations reported as absent from existing resources [59].

MDKG shows how an LLM can reduce annotation effort without replacing independent validation. It attaches contextual metadata to each relation instead of forcing every finding into an unconditional triple, and it evaluates downstream utility separately from extraction quality. Its scale also carries a caution: "novel" means not found in the comparison resources, not experimentally validated. New graph edges are hypotheses or extracted claims until corroborated.

C. LLM-assisted patient and task-specific graph construction

GraphCare uses LLM output together with external biomedical KGs to assemble patient-specific concept graphs, then applies a bi-attention GNN for mortality, readmission, length-of-stay, and medication prediction. It reported large gains on MIMIC-III and MIMIC-IV, especially in low-data settings [60]. Here the LLM acts as an open-world knowledge source for graph construction, while the downstream predictor is not itself an LLM. The case belongs in the survey because it shows how generated graph content can shape a clinical model even without natural-language generation at inference.

Patient-specific graphs improve relevance and complicate validation. The graph depends jointly on coding quality, patient-history completeness, LLM extraction, external KG coverage, and graph composition rules. Random train/test splits can leak recurring codes or graph neighborhoods. Temporal patient splits, source-specific ablations, and audits of generated edges are needed before a gain can be attributed to personalization.

D. Completion and semantic enrichment

Completion methods use LLM semantics to predict missing relations or improve graph representations. DrKGC learns structural embeddings and logical rules with a lightweight model, then retrieves a query-specific subgraph bottom-up and refines it with a GCN adapter. The structural signal is integrated into LLM fine-tuning. It reported improvements on general and biomedical completion datasets [61]. Its dynamic retrieval is more structure-aware than converting a fixed neighborhood to text, but predicted triples remain inductive outputs. Evaluation on randomly removed edges can overstate scientific discovery because neighboring structure may encode the answer, so temporal and source-held-out splits are stronger tests.

LLaDR prompts an LLM for treatment-oriented textual descriptions of biomedical concepts and uses those descriptions to fine-tune KG embeddings for drug repurposing [62]. The approach can enrich sparse entities with mechanistic context that the KG does not represent explicitly. That "common sense" may also be memorized literature, unsupported inference, or outdated practice. Description provenance and counterfactual tests are essential, especially when a repurposing candidate is justified with text generated by the same model that helped rank it.

MedCo, discussed above, combines EHR-mined associations, type-constrained LLM relations, generated descriptions and rationales, and heterogeneous graph learning [52]. Together, DrKGC, LLaDR, and MedCo mark a shift from using language only as node attributes toward using LLMs to synthesize relation semantics and training signals. They also make component attribution harder. At a minimum, studies should compare graph structure alone, verified source text alone, raw LLM descriptions, descriptions with source grounding, and the fused model.

E. Construction quality as a release process

The methods above motivate a four-state release model. A candidate is *extracted* when the LLM emits a schema-valid triple linked to a source span. It becomes *normalized* when entities and predicates map to controlled identifiers and qualifiers. It becomes *verified* when automated constraints and an independent model or curator confirm the evidence mapping. It becomes *asserted* only when an authorized curator approves it for a versioned production graph. Confidence scores do not replace these states; model probability, self-consistency, agreement among LLMs, source quality, and human approval measure different properties and should be stored separately. The graph must also support retraction, supersession, and rollback, or an automated updater can add claims without being able to correct them safely.

VII. Bidirectional, Agentic, Multimodal, and Temporal Systems

The families above treat construction and grounded generation as separate directions. Recent systems are increasingly bidirectional. A model extracts or proposes graph content, the graph guides later retrieval, and new evidence can trigger another construction cycle. The loop is attractive because biomedical knowledge is incomplete and moves fast. It is also where a useful retrieval component quietly becomes a knowledge-management system. Errors now persist across sessions, generated claims can pass for external evidence, and updates can change behavior without a release. Preserving the origin and status of everything exchanged between the model and graph is therefore the governing requirement.

A. Bidirectional co-augmentation

DALK is an early biomedical example. It uses a model to build an evolving Alzheimer's-disease graph from the literature, then brings selected graph evidence back into inference through coarse-to-fine sampling [63]. This addresses a genuine limitation of static graphs, since specialized evidence is often too recent or too granular to be there. It also exposes an ambiguity in the word "bidirectional," which covers two very different claims. One is an engineering claim: construction output becomes retrieval input. The other is an epistemic claim: the resulting graph is reliable enough to constrain answers. Only the first follows automatically. The second needs source-span grounding, predicate validation, versioned snapshots, and testing against evidence the same model did not produce.

Generate-on-Graph, though evaluated outside biomedicine, offers a useful abstraction. Its loop lets the model explore the graph and synthesize missing triples when the graph alone cannot complete a path [64]. Synthesis like this can raise recall. But a generated triple is an inference, not retrieved evidence. In a clinical deployment it belongs in a scratch graph scoped to the current request. It may support a hypothesis or shape a search query. It must not inherit the status of a curated edge merely because it completed a plausible path.

KGARevion runs a safer asymmetric loop. The model proposes candidate triples, and the graph verifies and filters them before an answer is written [54]. The model supplies recall, the graph keeps precision. This works well when the graph is authoritative in its domain. One ambiguity remains: "not found" can mean the claim is false, poorly normalized, out of scope, or simply absent from this release. Verification should distinguish these cases. A typed verdict of supported, contradicted, unresolved, or out of scope is more useful than collapsing absence into falsity.

Table III sorts these systems by what actually crosses the feedback boundary. The distinction heads off a common error: a system can be bidirectional at inference without being allowed to write to the institutional graph.

TABLE III
Bidirectional systems, distinguished by what actually crosses back into the graph. "Bidirectional" is often used as a single label, but the loops differ in the one respect that matters for safety: how long the model's output persists and what authority it acquires. A hypothetical edge invented to complete a reasoning path should live only for the current request. A candidate relation extracted from a paper may sit in a staging area awaiting review. Only a curator-approved assertion belongs in the graph that later queries treat as authoritative. The rightmost column is therefore a design requirement rather than a description: a system that lets a reasoning-loop artifact acquire maintenance-loop persistence is the circular self-validation failure of Section XII.

Loop type	LLM-to-graph / graph-to-LLM artifact	Persistence
Reasoning	Hypothetical edge; neighbor/path evidence	Request-scoped
Retrieval	Query reformulation; source-linked subgraph	Logged, not asserted
Curation	Candidate relation; constraint and curator feedback	Candidate store
Learning	Synthetic instruction/reward; graph supervision	Versioned artifact
Maintenance	Proposed addition/correction; existing claims and provenance	Quarantined until approval

B. When is a system genuinely agentic?

The word *agentic* gets attached to any multistage prompt chain. We reserve it for something narrower: a controller that watches an evolving state, picks among available actions, judges what comes back, and decides whether to continue, revise, abstain, or stop. A fixed sequence of link, retrieve, generate is a pipeline, even when every stage calls a model. An agent might broaden a subgraph, fetch the source article, issue a structured query, test a contradiction, or ask for human review, depending on what it found. The distinction is not pedantry. A pipeline's failure modes can be listed in advance. An agent's depend on trajectories that only exist at run time.

C. The tool surface over a medical graph

In practice an agent is defined by the tools it can call, which makes the tool surface the main design artifact. Six operations are worth separating, because each carries a different risk.

Concept resolution maps a mention to identifiers. It should return ranked candidates with confidence, not one answer, because an early commitment to the wrong identifier constrains everything downstream. *Neighborhood expansion* retrieves typed neighbors under an explicit hop and degree budget; without a budget, one hub concept exhausts the context window. *Path query* finds routes between anchor concepts under relation-type constraints. *Structured query execution* runs a generated SPARQL or Cypher statement, read-only, against an allowlisted grammar. *Source retrieval* fetches the documents behind an edge, which is how an agent recovers the dose, population, and study design a triple cannot carry. *Verification* checks a claim against the graph.

Verification should return a typed verdict rather than a boolean, because collapsing absence into falsity turns a graph-coverage limit into an apparent factual finding [54]. Write access also belongs on a separate tool with its own authorization.

An agent that can propose an edge and an agent that can assert one are different risk classes, which Section XII takes up.

D. Planning, control, and stopping

General graph agents supply traversal patterns that biomedical systems inherit. Think-on-Graph runs an iterative beam search, letting the model pick promising relations and entities [65]. Reasoning on Graphs separates planning from execution, generating relation-path plans grounded in the graph before any answer [66]. This planning–execution separation transfers well to biomedical systems. A relation-path plan can be checked against the schema before retrieval runs, catching schema-incompatible plans early. Think-on-Graph 2.0 alternates graph and document retrieval, so structure deepens the document search while text refines the graph exploration [67]. The alternation suits biomedicine well. The graph finds the relational route; the documents supply the dose, population, and study design that no triple carries.

Control is where biomedical agents usually go wrong. Every extra action widens the attack surface and leaves more room for confirmation bias. An agent that keeps expanding until it finds support will always find some. Three controls belong in any implementation. Cap the tool calls and the cumulative retrieved context. Require a search for disconfirming evidence, not just corroboration. Expose the stopping criterion, whether that is evidence sufficiency, a contradiction threshold, a budget, or an abstention. HyKGE shows the hazard from the retrieval side, since a provisional hypothesis used to guide exploration can steer the search toward paths that agree with it [41]. An agent that writes its own hypotheses has the same problem and more iterations in which to compound it.

E. Multi-agent decomposition

Splitting a task across specialized agents has the strongest clinical evidence behind it. The oncology trial-matching platform separated extraction from eligibility reasoning. Collapsing the two into one call lowered extraction fidelity and raised unresolved conflicts, dropping F1 from 0.82 to 0.78 [68]. That is a rare direct measurement of a decomposition benefit. The mechanism is worth naming: keeping extraction and reasoning separate keeps each stage inspectable, and stops an error in one from being laundered through the other.

The same study shows the complementary control. Its final eligibility verdict came from deterministic rules rather than a generator, followed by clinician review. CrossDDI makes the same argument for drug interactions, taking the model out of arbitration entirely and issuing a positive verdict only on explicit supporting evidence, which cut variation across three backbones [55]. For any decision expressible as a rule, the defensible design is agents for gathering and a deterministic component for deciding. This tradeoff reduces coverage while improving traceability and stability across model versions.

F. Agents that maintain the graph

AMG-RAG is the clearest case of an agent whose job is the graph rather than the answer. It automates construction and continuous updating, and retrieves current graph evidence for question answering [69]. MedKGent scales the ingestion side, using two agents to extract and integrate relations from over ten million PubMed abstracts while resolving conflicting claims [70].

Benchmark accuracy is the wrong endpoint for both. A wrong answer is a visible, bounded error. A wrong persistent edge is neither. It contaminates every later retrieval, training set, explanation, and decision that touches it, silently. Update evaluation needs its own metrics: candidate-edge precision by relation type, conflict-resolution accuracy when sources disagree, time from published correction to graph correction, how far a bad edge propagates, and whether rollback actually works. None of these is reported anywhere in the current literature. That gap matters more than any remaining headroom on question-answering accuracy.

G. What an auditable agent trace contains

Reproducing an agent run takes more than the prompt. An auditable trace should record the initial request and policy context, each state representation, the chosen action and available alternatives, and exact tool inputs and outputs. It should also identify the model, graph, ontology, retriever, and prompt versions; source timestamps; evidence retrieved and rejected; confidence or abstention decisions; and the terminal reason. This is not equivalent to publishing hidden chain-of-thought, which may be incomplete, unstable across runs, or disclosive. A concise event log captures the externally meaningful decisions without storing unrestricted internal reasoning and makes a failure attributable after the fact. Reproducibility also requires a snapshot mode, so the same request can be replayed against frozen dependencies even while the production graph continues to update.

H. Multimodal graphs as an alignment layer

Biomedical entities are described by text, images, molecular structures, sequences, physiological signals, and clinical events, and a graph can align them. Multimodal graph integration takes three distinct forms. The first aligns encoders through graph relations, as in BioBridge [50] and the sequence-augmented PrimeKG++ resource [71]. The second stores multimodal evidence as graph attributes for retrieval at inference. The third uses graph-derived supervision to train a multimodal model, as in MedKInstruct [53]. The first two are graph–foundation-model integrations rather than generative systems, so their endpoint is cross-modal retrieval quality and inductive performance on unseen entities, not answer faithfulness. The third relocates the graph from context to supervision, which improves compositional reasoning and makes leakage harder to see: a test answer or near-duplicate caption present during instruction synthesis lets the model reproduce graph-to-label mappings without attending to the image. Image-counterfactual tests and entity-disjoint splits are the minimum defense.

Physiological signals are the newest frontier. EEG-MedRAG organizes domain knowledge, patient cases, and a repository into a three-layer hypergraph for joint semantic and temporal retrieval [72]. This architecture can represent the n-ary clinical relations that plain triples flatten, but its clinical value requires prospective testing across devices and institutions. Across all three designs, a multimodal system should report how it behaves when a modality is missing, how calibrated it stays when modalities disagree, and whether a given explanation rests on the patient signal or on generic graph knowledge.

I. Temporal knowledge is more than a fresh graph

Biomedical claims run on four clocks, and conflating them is a common design error. Transaction time records when an edge entered a database, publication time when the supporting evidence appeared, valid time when a recommendation actually applies, and patient-event time when something happened to a specific person. A graph rebuilt every night has excellent transaction-time resolution and can still be temporally naive, because none of that tells you whether a recommendation was superseded.

MedKGent shows the ingestion side is tractable, using two agents to extract and integrate relations from more than ten million PubMed abstracts spanning several decades and building a daily series of graphs while resolving conflicting claims [70]. A series of snapshots, though, captures publication and transaction history rather than clinical time. What clinical reasoning needs are statements of the form "this symptom typically precedes that event," or "this therapy was contraindicated during that interval." ChronoMedKG is the first resource to attempt them at scale, reporting 460,497 evidence-linked triples over 13,431 diseases with onset windows and traceable support, alongside a temporal question benchmark with static controls; performance drops from static to temporal questions and graph retrieval recovers only part of it [13]. The work was a preprint at the time of writing, so its numbers are indicative rather than settled, but the direction is hard to dismiss.

The evaluation requirement that follows is an as-of boundary. For a question asked at time t, every edge, document, index, adaptation set, and cached summary must be restricted to what existed by t. Random edge splits leak future knowledge even when the visible prompt contains only older facts, and a contemporary model leaks more. Strong evaluations add evolving-guideline cases, retractions, and adversarial conflicts between older high-volume evidence and newer authoritative guidance. A system should also be explicit about which question it answers: what is believed now, what was known then, or what applied to this patient at the time. These are not interchangeable.

J. Preventing circular self-validation

Closed-loop systems create a distinctive failure mode. Suppose model M proposes claim c, a graph updater stores c, and a later instance of M retrieves that edge as "external" support. The loop has converted one model output into apparently independent evidence. Self-consistency, repeated extraction, or agreement among closely related models can amplify the effect without adding a genuinely independent source. The resulting citation can be syntactically correct and epistemically circular.

Four controls are mandatory. Maintain separate asserted, candidate, and ephemeral graph zones, and let only asserted edges serve as authoritative grounding. Require every persistent claim to point to a resolvable primary or curated source span, with model identity recorded as an extractor rather than an evidence source. Require independent validation before promotion, by a curator, a deterministic database check, an independently sourced document, or a suitably independent model with calibrated disagreement handling. Make promotion, supersession, retraction, and rollback explicit, permissioned events. Agent tools should hold least privilege: retrieval agents get read access, extraction agents may write candidates, and only an authorized release process may write asserted edges.

The same separation must hold through generation. A model-generated summary is a derived artifact, not a primary source; a graph-consistent statement is not necessarily entailed by the cited document; and a source-supported sentence can still be clinically inapplicable because of a population or temporal mismatch. As the sentence-level analysis in VERICITE shows, supplying relevant documents does not by itself ensure faithful claim-citation alignment [11]. Trustworthy closed-loop systems must preserve lineage from answer claim to graph assertion to source passage, expose uncertainty and conflicts, and abstain when the chain is incomplete. Feedback may improve coverage, but only independently governed evidence may increase authority.

VIII. Data Resources, Ontologies, Clinical Data, and Benchmarks

The behavior of a biomedical KG–LLM system is bounded by the knowledge substrate it runs on. Graph size is a poor proxy for fitness. Two graphs with similar triple counts can differ sharply in semantic precision, provenance, temporal coverage, licensing, and fit for a target task. A drug–disease edge can denote an approved indication, an off-label use, a contraindication, a statistical association, a text-mined co-mention, or a model-generated hypothesis. Flatten those into one edge vocabulary and the downstream LLM receives apparent structure with no defensible evidentiary semantics. The relevant unit of analysis is therefore not the graph alone but the tuple

$$\langle \text{snapshot, schema, provenance, retrieval, serialization, model} \rangle. \tag{2}$$

This matters more as biomedical KGs have grown from curated molecular networks into multimodal, literature-scale resources. PrimeKG integrates 20 resources into 4,050,249 relations across ten biological scales and separates indications, contraindications, and off-label uses [4]. SPOKE integrates 41 databases through 21 node and 55 edge types [2], while the Clinical Knowledge Graph connects public databases, literature, and patient proteomics in roughly 20 million nodes and 220 million relationships [3]. At a different scale, PubMed Knowledge Graph 2.0 connects more than 36 million papers, 1.3 million patents, and 0.48 million clinical trials through biomedical entities, citations, projects, and author networks [6]. These are not interchangeable stores: they encode different objects, different evidence, and different notions of relation validity.

A. Representative knowledge resources

Fig. 6 places the representative resources by how each was built and what scale it describes, and Table IV gives their properties and the roles each best supports. The layout avoids a single "largest graph" ranking, because the constraints that matter for graph-grounded generation, such as evidence traceability, update cadence, and terminology alignment, do not reduce to scale.

Several implications follow. Integrated KGs should preserve the identity of each contributing source and the method by which each edge was obtained, because a curated assertion, a statistical association, a single-abstract extraction, and an LLM-proposed edge deserve different default trust. An evaluation must name the exact snapshot, not just the resource, since PrimeKG construction scripts and source releases have changed after publication and terminologies issue regular versions; silently using a later graph breaks reproducibility or leaks post-cutoff knowledge. Coverage must be measured against the target cohort and task, because global node or edge counts hide whether the decisive entities, synonyms, relations, and temporal qualifiers are present.

B. Ontologies and terminologies as semantic contracts

Biomedical KG–LLM systems often treat UMLS, SNOMED CT, ICD, RxNorm, LOINC, HPO, MONDO, the Gene Ontology, ChEBI, UniProt, and Reactome as interchangeable "knowledge bases." They do different jobs. UMLS is chiefly a metathesaurus and mapping infrastructure [21]; SNOMED CT is a compositional clinical terminology [22]; ICD supports classification and reporting; RxNorm normalizes clinical drugs [23]; LOINC identifies observations [24]; HPO represents phenotypic abnormalities [25]; MONDO harmonizes disease concepts [26]; and the Gene Ontology, ChEBI, UniProt, and Reactome describe functions, chemicals, proteins, and pathways [27, 28]. Their value lies as much in constraining entity identity and relation semantics as in adding content.

The survey distinguishes four ontology-facing operations: normalization (mapping mentions to identifiers), semantic expansion (retrieving synonyms or descendants), constraint enforcement (validating entity and relation types), and reasoning (using asserted or inferred hierarchies). Conflating them hides failure propagation. A system can answer incorrectly because the LLM reasoned poorly, because it linked "MS" to mitral stenosis rather than multiple sclerosis, because it expanded an ICD billing code too broadly, or because it traversed a relation whose direction was lost during textualization.

Licensing is part of system design, not an administrative afterthought. UMLS access requires accepting its license and reporting a release identifier [77]. SNOMED CT is free in member territories but stays subject to affiliate and national conditions, with fees or exemptions elsewhere [78]. A paper claiming an "open" pipeline should separate open source code from redistributable graph content, and should report terminology edition, country extension, language edition, mapping version, and whether released prompts or artifacts reproduce licensed strings.

C. Clinical data and patient-specific graphs

Patient graphs create a second, more demanding regime. EHR-derived nodes and edges are observations made through a workflow, not direct measurements of an underlying biological state. A diagnosis code may reflect rule-out reasoning, reimbursement, history, or active disease; a medication order differs from administration; and a discharge summary is written after many outcomes are known. Building an admission-time prediction graph from a completed encounter can therefore introduce post-index leakage even when the patient split is correct.

MIMIC-IV shows both the value and the limits of public clinical data. It links structured hospital and intensive-care data with separately released notes and imaging, but it reflects routine care at one Boston academic center and carries known archival idiosyncrasies [79]. Its note resource is credentialed, governed by a data-use agreement, and contains 331,794 discharge summaries and more than 2.3 million radiology reports [76]. A KG–LLM evaluation on MIMIC should specify the database and module versions, cohort SQL, index time, censoring rules, code mapping, handling of repeated admissions, and whether graph edges came from information documented after the prediction time.

The unit of splitting must follow the intended deployment question. Multiple encounters from one patient must not cross train and test partitions unless the task is explicitly within-patient forecasting. Temporal evaluation should train on earlier care and test on later care, and institutional transport should be tested on a second health system or a deliberately shifted dataset. Deidentification does not license unrestricted prompt transmission: a study using hosted models must disclose which data fields left the secure environment, the applicable terms, and whether the provider retained prompts.

D. A minimum data card for KG–LLM studies

Two nominally identical "PrimeKG plus GPT" experiments can differ enough that their results are not comparable, and the difference usually hides in unreported detail: which snapshot, which terminology release, which edges were curated versus text-mined, what the unresolved-mention rate was. A short data card makes those differences visible, and the field list we recommend is given in the supplementary material [20]. Anything that changes what the graph can answer belongs on the card. Provenance, temporal fields, and coverage over the evaluated entities matter more than aggregate size.

E. Tasks, datasets, and benchmarks

No single benchmark establishes the quality of a biomedical KG–LLM system, and the temptation to report one headline number is what allows several distinct failures to hide. Evaluation should span a stack aligned with the pipeline:

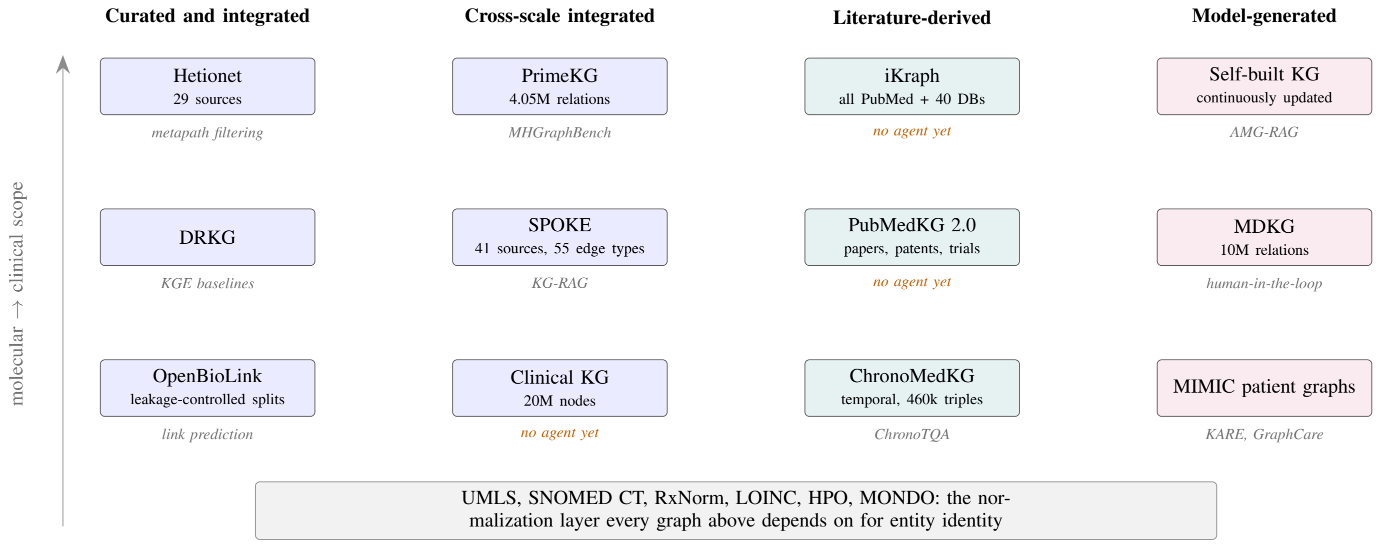

Fig. 6. The representative resources, grouped by how each graph was built and ordered downward by scale from molecular to clinical, with the systems built on each shown beneath it. Entries marked in orange have no agentic system in the reviewed literature despite carrying the provenance and temporal metadata an agent needs. They therefore represent a clear near-term opportunity. The terminologies are drawn as a band rather than as separate nodes because every graph above them depends on that layer for entity identity. A terminology error can therefore propagate through every dependent graph.

TABLE IV

THE REPRESENTATIVE BIOMEDICAL KNOWLEDGE GRAPHS, WHAT EACH IS GOOD FOR, AND WHAT TO REPORT WHEN USING IT. THE RESOURCES DIFFER IN HOW EACH EDGE CAME TO EXIST, WHETHER AN ASSERTION CAN BE TRACED TO A SOURCE, HOW OFTEN THE GRAPH IS REFRESHED, AND WHICH SEMANTIC DISTINCTIONS THE SCHEMA PRESERVES. A GRAPH THAT COLLAPSES INDICATION, CONTRAINDICATION, AND OFF-LABEL USE INTO A GENERIC DRUG–DISEASE RELATION CANNOT SUPPORT CONTRAINDICATION-AWARE RETRIEVAL. THE FINAL COLUMN LISTS THE LIMITATIONS THAT A PAPER USING EACH RESOURCE SHOULD STATE. OMITTING THEM MAKES TWO NOMINALLY IDENTICAL EXPERIMENTS INCOMPARABLE.

Resource	Content and value	Suitable KG–LLM roles	Principal limitations
PrimeKG [4]	Precision-medicine graph; 20 sources, >4M relations across ten scales	Disease-centered retrieval, path reasoning, drug repurposing, verification	Static snapshots; provenance varies by source; a graph edge is not a clinically actionable fact
Hetionet / OpenBioLink [1, 73]	Heterogeneous repurposing network; leakage-controlled link-prediction splits	KGE/GNN baselines, link prediction, structure ablation	Discovery-oriented; link prediction is not clinical validation; random splits overstate utility
SPOKE [2]	Broad integrative graph, 41 sources, molecular to clinical	Cross-scale retrieval, cohort phenotyping, patient linkage	Complex source heterogeneity; reproducible snapshots need careful release management
Clinical Knowledge Graph [3]	Proteomics, databases, ontologies, literature	Multi-omics interpretation, experiment-centric querying	Proteomics-centered; curated and derived evidence must stay distinguishable
DRKG [74]	Drug-repurposing graph with KGE tools	Drug–disease link prediction, KGE/GNN comparison	Provenance, inverse edges, negatives, snapshot date drive evaluation; links are hypotheses
iKraph [5]	Literature-scale graph from all PubMed abstracts plus 40 databases	Literature mining, hypothesis generation, provenance-aware discovery	Extraction errors propagate; abstract-level evidence omits full-text qualifiers
PubMedKG 2.0 [6]	Papers, patents, trials, bioentities, citations, projects	Translational evidence tracing, trial-aware RAG	Scholarly-output graph, not a clinical-fact graph; linkage is not evidentiary support
PrimeKG++ / BioMedKG [71]	PrimeKG with text and biological sequences	Graph–text–sequence fusion, multimodal link prediction	Random edge splits do not establish temporal generalization; gains may depend on node attributes
MIMIC-IV patient graphs [75, 76]	Diagnoses, meds, procedures, labs, notes, imaging links	EHR prediction, longitudinal retrieval, patient-specific explanation	Single-center, retrospective, access-restricted; post-outcome leakage risk

construction, normalization, relation extraction, completion, retrieval, structured reasoning, generation, evidence attribution, and downstream use. A final-answer QA score can conceal a failed retriever rescued by parametric memory. A high link-prediction score can be driven by inverse relations or graph redundancy. A correct evidence path can still produce an unsafe response if the generator omits a contraindication or declines to abstain. The supplementary material [20] maps the common benchmarks against what each does and does not establish.

Three mismatches recur across otherwise careful papers. *Task mismatch* arises when exam-style QA is used to support a claim about clinical diagnosis, or link prediction to support one about drug efficacy. *Knowledge mismatch* arises when the questions were generated from the same graph supplied at inference, which makes agreement with that graph nearly automatic and says little about correctness beyond it. *Interface mismatch* arises when graph facts are serialized in a format that happens to suit one model. KG-LLM-Bench, a workshop preprint, reported that textualization is itself an experimental factor rather than an implementation detail, and MHGraph-Bench found in peer-reviewed work that short graph snippets improved some models and degraded others [80, 12]. A result reported under one serialization does not transfer.

F. Leakage and benchmark validity

Leakage in KG–LLM studies happens at more points than ordinary train-test contamination, and it should be treated as a threat to the claimed graph contribution rather than as a hygiene issue. Beyond parametric contamination of public benchmarks, the graph itself opens routes: questions or gold paths generated from the inference graph, test edges recoverable from their inverses, snapshots that postdate the target event, patient encounters split across partitions, labels encoded in ontology definitions, and the same model family constructing, answering, and judging. The supplementary material [20] tabulates each mechanism with its control.

The concern is empirical, not procedural. In a poisoning study, models trained with medically harmful misinformation kept performance comparable to unpoisoned controls across common medical benchmarks, so the benchmark scores missed a clinically important degradation entirely [81]. The same study found a graph-based screen detected 91.9% of harmful passages at 80.3% precision, which shows both the value and the boundary of a curated graph as an independent constraint. Leakage also distorts graph-native evaluation: removing train-test redundancies changed embedding-model performance across several biomedical graphs, and results on a real-world rare-disease repurposing set were far worse than on indications sampled at random from the graph [82]. That was preprint evidence, so the estimates are indicative, but the design principle is not in doubt. A random held-out edge does not simulate future discovery.

IX. From Resources to Systems: What Each Graph Enables

Section VIII described the graphs as data. This section asks the question an engineer actually starts with: given a particular graph and a particular clinical or scientific task, what can be built on it, what has already been built, and what does its structure make possible that a flat text index does not? The framing matters because graph choice is not a neutral preliminary. It fixes which retrieval mechanisms are expressible, which agent actions are meaningful, and which failure modes the system inherits.

A. Structure determines the retrieval mechanism

Modern retrieval-augmented generation offers a standard toolkit: dense passage retrieval, sparse lexical matching, hybrid fusion, cross-encoder reranking, query decomposition, hypothetical document expansion, and iterative or corrective loops that re-retrieve when the first attempt looks insufficient. Applied to a graph, each of these behaves differently depending on what the graph actually encodes, and the difference is not cosmetic.

Typed edges make relation-constrained retrieval expressible. SPOKE distinguishes 21 node and 55 edge types [2]. A query that needs "drugs that target a protein in this pathway but are not contraindicated in renal impairment" is a typed traversal with a negative constraint. A dense index over verbalized triples cannot express the constraint; it can only hope that semantic similarity approximates it. The engineering consequence is that graphs with rich edge typing reward symbolic or hybrid retrieval, while schema-light graphs push you toward embedding similarity and therefore toward all of its failure modes.

Evidence-bearing edges make provenance-first ranking possible. iKraph carries relations from all PubMed abstracts alongside database-derived assertions and probabilistic inference [5], and MDKG attaches conditional statements, cohort characteristics, source, and confidence to each relation [59]. On such a graph, retrieval can rank by evidence strength rather than by embedding proximity, and it can return supporting and contradicting evidence as separate result sets. On a graph without assertion-level provenance, neither is available, and a system built there cannot honestly implement claim-level attribution no matter how good its generator is.

Semantic distinctions in the schema make safety-relevant retrieval possible. PrimeKG separates indications, contraindications, and off-label uses [4]. That single design choice is what allows a retriever to be evaluated on contraindication recall, which for a prescribing-support task matters more than aggregate accuracy. A graph that collapses these into a generic *drug–disease* relation cannot support the evaluation, so the safety property becomes unmeasurable rather than merely unmet.

Hierarchy and community structure make global questions tractable. Path retrieval answers local, relation-specific questions well and broad synthesis questions poorly. Community detection over a large graph, following the pattern established for query-focused summarization [43], gives a retriever something to return when the question has no single decisive path. KARE exploits exactly this on a multi-source concept graph [44], and MedRAG builds a purpose-shaped four-tier hierarchy so that retrieval discriminates among diseases with overlapping presentations rather than merely finding similar text [45].

Attribute-rich nodes make cross-modal retrieval possible. PrimeKG++ and BioMedKG add textual descriptions, SMILES strings, and amino-acid and nucleotide sequences

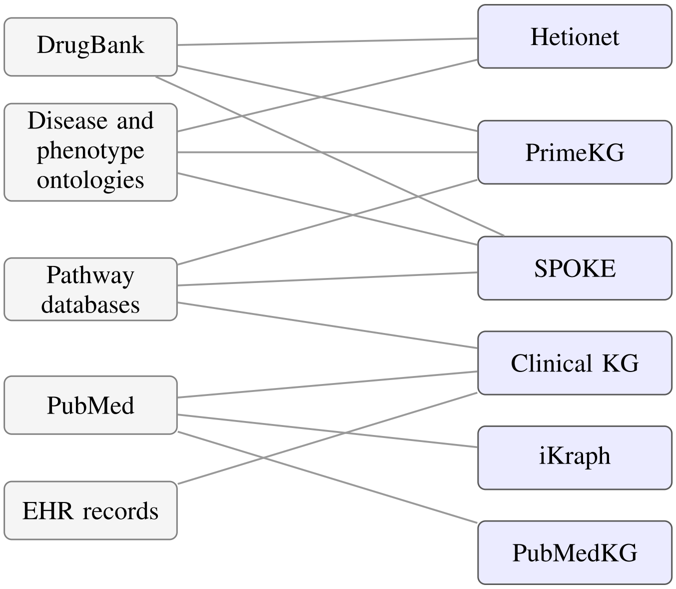

Fig. 7. Shared provenance among the representative graphs. Because several draw on the same upstream databases, two evaluations that use different graphs are not necessarily independent: an error or bias in DrugBank or in a disease ontology propagates into Hetionet, PrimeKG, and SPOKE alike. A result replicated across graphs that share sources is weaker evidence than the count of graphs suggests.

to graph nodes [71]. This turns the graph into an alignment substrate: retrieval can cross from a molecule to a protein to a disease phenotype through learned transformations rather than through text alone, which is the mechanism BioBridge formalizes [50].

B. *What has been built, and on what*

Table V joins graphs to the systems built on them, the retrieval mechanism each uses, and the agentic capability each demonstrates. Several widely cited graphs have no agentic system built on them despite carrying the metadata such a system needs. A number of systems also omit the graph snapshot they used. The table records that omission rather than inferring a snapshot; unreported provenance is a finding about the literature, not a gap in the table.

C. *Where the opportunities sit*

Reading the table by its empty cells is more informative than reading it by its entries. Three gaps stand out, and one caution applies to all of them: several of these graphs share upstream sources, as Fig. 7 shows, so results replicated across them are less independent than a count of graphs implies.

The large literature-scale and multi-omics graphs, iKraph, PubMedKG 2.0, and the Clinical Knowledge Graph, carry exactly the properties an agent needs, namely resolvable provenance, temporal metadata, and links between assertions and the documents or experiments that support them, and yet no agentic system in the reviewed literature is built on them. An agent over PubMedKG 2.0 could trace a claim from a guideline to the trials that support it to the patents that surround it, which is a translational-evidence workflow no current system performs. The obstacle is engineering effort rather than any missing capability.

Contraindication-aware retrieval is expressible on PrimeKG and is not being evaluated. The graph makes the distinction available, yet the reviewed systems still report aggregate accuracy. A safety property that the graph makes directly testable therefore remains unevaluated because the evaluation convention has not caught up with the data.

Deterministic arbitration over agent output has one strong precedent [68] and one clear architectural argument in its favor [55], and it remains rare. For any task where eligibility, dosing, or interaction can be expressed as a rule, letting agents gather evidence and letting a deterministic component decide is both more auditable and more stable across model versions than asking a generator to arbitrate.

D. *Representative systems side by side*

Table VI places twelve representative systems on the dimensions a practitioner must specify: the graph and its type, the coupling depth of Section IV, the target application, the advantage as the authors report it, and the limitation that bounds the claim. Two patterns in the table matter more than any individual row.

First, the backbone column is almost empty. Of these twelve systems, none reports the language model and version behind its headline result in a form that would let a reader reproduce it; the only named models in the whole set are the GPT-4 baselines that the trial-matching study measured against. This is a property of the literature, not an omission in this survey. It is also the reporting failure that Section XII asks deployed systems to avoid. A graph-augmented result stated without its backbone cannot be reproduced, because the same graph in front of a stronger model is a different experiment, and the shrinking-gain effect discussed in Section III makes that difference decisive rather than incidental.

Second, coupling depth clusters at C1 and C2. Eleven of the twelve keep the graph outside the weights, which preserves auditability and in-place correction. Only the trial-matching platform reaches C5, and it is also the only one of the twelve with prospective clinician-facing evaluation. Whether that association is causal or an artifact of who builds deployable systems cannot be settled from twelve cases, but it is the clearest signal in the table.

E. *Choosing a configuration for a task*

The preceding analysis supports a short design path. Start from the question type. Relation-specific and mechanistic questions call for typed path retrieval on a schema-rich graph, which is coupling C1 with a symbolic component. Broad synthesis and patient-context enrichment suit community or hierarchical retrieval. Questions turning on qualifiers, dose, population, or study design call for hybrid graph-and-text retrieval, because a triple will not carry those qualifiers while a passage can. Exact constraints suit symbolic query execution with the model confined to translating the question, and eligibility or interaction decisions require a deterministic arbiter

TABLE V

WHICH SYSTEMS HAVE BEEN BUILT ON WHICH GRAPH, AND WHAT EACH GRAPH WOULD ADDITIONALLY SUPPORT. THIS TABLE JOINS THE RESOURCES OF SECTION VIII TO THE METHODS OF SECTIONS V ONWARD, SO THAT A READER STARTING FROM A CHOSEN GRAPH CAN FIND THE CLOSEST PRIOR WORK. "AGENTIC CAPABILITY" MEANS THE SYSTEM CHOOSES AMONG ACTIONS BASED ON INTERMEDIATE RESULTS RATHER THAN FOLLOWING A FIXED SEQUENCE; A PIPELINE THAT ALWAYS LINKS, RETRIEVES, THEN GENERATES IS NOT AGENTIC EVEN WHEN EVERY STAGE CALLS A MODEL. "NOT REPORTED" RECORDS THAT THE SOURCE PAPER DOES NOT STATE ITS GRAPH SNAPSHOT, WHICH IS ITSELF A FINDING ABOUT THE LITERATURE. THE EMPTY AND ORANGE CELLS ARE THE MOST INFORMATIVE PART: THE FINAL ROW LISTS THREE WIDELY USED GRAPHS THAT CARRY THE PROVENANCE AND TEMPORAL METADATA AN AGENT NEEDS AND ON WHICH NO AGENTIC SYSTEM HAS BEEN BUILT.

Graph	Systems built on it	Retrieval mechanism	Agentic capability	What the structure additionally unlocks
SPOKE [2]	KG-RAG [33]	Prompt-aware context extraction with embedding-based pruning	None; single-pass retrieval	Typed multi-hop traversal with negative constraints; cross-scale queries from variant to phenotype to drug
UMLS [21]	DR.KNOWS [39], medIKAL [40], ILlama [35]	Path ranking via GNN and path encoder; typed entity weighting; precomputed triple documents	Iterative path expansion toward diagnosis concepts	Concept normalization as a shared interface across sites; assertion-status-aware patient graphs
PrimeKG [4]	MHGraphBench tasks [12]; controlled retrieval study [32]	Snippet injection; counterfactual-controlled retrieval	None reported	Contraindication-aware retrieval and evaluation; ten-scale reasoning; drug repurposing with explicit indication semantics
Hetionet / DRKG [1, 74]	Metapath filtering [83]	Task-driven metapath selection	None; graph-native prediction	Learned metapath priors as an agent tool; hypothesis ranking with leakage-controlled splits
Multi-source concept graph	KARE [44]	Hierarchical community retrieval plus patient-context augmentation	Reasoning-trace distillation, not run-time agency	Community summaries as an agent's coarse index before path-level drill-down
Four-tier diagnostic hierarchy	MedRAG [45]	Hierarchical discrimination plus similar-EHR retrieval	Follow-up question generation	Differential-diagnosis narrowing as an explicit agent objective
User docs plus authoritative sources	MedGraphRAG [46]	Top-down precise retrieval with bottom-up refinement	Two-stage refinement loop	Claim-level entailment checking against the linked source layer
AlzKB	ESCARGOT [84]	Executable graph-of-thought with graph queries	Plan, execute, inspect over a disease graph	Template for disease-specific agentic discovery on other curated graphs
Chemical reaction KG	KGRxn-LLM [42]	Agentic retrieval over a four-level hierarchy	Full agent loop with covered/holdout evaluation	The clearest existing model for separating graph-covered from generalization performance
Oncology KG (proprietary)	Trial matching [68]	Ontology normalization plus deterministic eligibility rules	Multi-agent extraction then reasoning, with clinician review	Deterministic arbitration over agent output; the strongest clinical-stage precedent
Self-built medical KG	AMG-RAG [69]	Continuous construction with current-evidence retrieval	Construction, update, and retrieval agents	Automated maintenance; also the largest untested risk surface
iKraph [5], PubMedKG 2.0 [6], CKG [3]	None identified	—	None	Provenance-ranked retrieval; trial-aware and patent-aware evidence tracing; multi-omics agent tools

rather than a generator, which is C5 with the verifier outside the model.

Then check three properties of the candidate graph before committing: whether it contains the decisive entities and relations for the target cohort, whether it carries assertion-level provenance if the output must be attributable, and whether its update cadence matches how fast the underlying knowledge moves. Inadequate coverage makes it the wrong graph for the cohort. Missing provenance forecloses claim-level attribution regardless of what the generator does. An update cadence that lags the domain leaves the system confidently wrong about recent evidence, and no retrieval mechanism repairs that.

X. Layered Evaluation and Comparative Evidence

KG–LLM systems should be evaluated as pipelines with observable intermediate states. Let G denote a versioned graph, $R(q, G)$ the retrieved subgraph or evidence set for query q, and $y = M(q, R)$ the generated output. Final-answer quality alone cannot say whether an error came from the graph, normalization, retrieval, reasoning, or generation, and a high-quality graph does not guarantee that the LLM uses it. We propose seven evaluation layers, summarized in Table VII.

The first five layers should appear even in an offline system paper. Layers 6 and 7 become mandatory as claims move from algorithmic performance to translational or clinical utility.

TABLE VI
TWELVE REPRESENTATIVE SYSTEMS ON THE DIMENSIONS A PRACTITIONER MUST SPECIFY. COUPLING DEPTH FOLLOWS THE C-LEVELS OF SECTION IV. "BACKBONE" RECORDS THE LANGUAGE MODEL AND VERSION AS REPORTED BY THE SOURCE PAPER; *n.r.* MEANS NOT REPORTED, WHICH IS THE VALUE FOR EVERY SYSTEM HERE AND IS ITSELF THE FINDING DISCUSSED IN THE TEXT. ADVANTAGES ARE AS CLAIMED BY THE AUTHORS, WITH FIGURES QUOTED ONLY WHERE THE SOURCE STATES THEM; LIMITATIONS ARE THE CONSTRAINTS THAT BOUND EACH CLAIM. EVIDENCE MATURITY FOLLOWS THE E-LEVELS OF SECTION XI-G.

System	Graph (type)	Coup.	Back.	Reported advantage	Bounding limitation
KG-RAG [33]	SPOKE (curated, cross-scale)	C1	n.r.	Cut token use by more than half with no loss on the tested tasks	Efficient evidence packaging, not evidence of symbolic reasoning
KRAGEN [38]	n.r.	C1	n.r.	Graph-of-thought decomposition with per-subproblem retrieval	Commits early; a wrong first plan cannot be recovered by later retrieval
DR.KNOWS [39]	UMLS (ontology)	C2	n.r.	Gains on MIMIC-III and an institutional dataset	Incorrect starting concepts a recurring error cause; E1
medIKAL [40]	UMLS (ontology)	C2	n.r.	Recovers diseases absent from retrieved paths	Blurs attribution between graph and initial guess; language-specific; E1
KARE [44]	Multi-source concept graph	C1–C2	n.r.	Gains on mortality and readmission, MIMIC-III/IV	Four mechanisms confounded; label-derived traces need a leakage check; E1
MedRAG [45]	Four-tier diagnostic hierarchy (purpose-built)	C2	n.r.	Discriminates diseases with overlapping presentations	Narrow by construction; private evaluation limits replication; E1
MedGraphRAG [46]	User documents plus authoritative sources	C2	n.r.	Nine QA benchmarks, two fact-checking sets, long-form generation	Benchmark-level only; no claim-level entailment or clinician study; E1
KGARevion [54]	Grounded domain graph (n.r.)	C5	n.r.	Parametric recall with graph-enforced authority	Ceiling is graph completeness; "verified" means present in this snapshot
ESCARGOT [84]	AlzKB (disease graph)	C2	n.r.	Gains over conventional RAG on open-ended questions	An inspectable trace is not a faithful scientific explanation
AMG-RAG [69]	Self-built, continuously updated	C5	n.r.	Automated construction and update with current-evidence retrieval	Largest untested risk surface; no update metrics reported
MedKGent [70]	Self-built from >10M PubMed abstracts	C5	n.r.	Daily graph series with conflict resolution across decades	Captures publication time, not clinical time; preprint
Oncology trial matching [68]	Oncology KG (proprietary)	C5	n.r.	F1 0.82 vs 0.47 GPT-4 zero-shot and 0.67 CoT; 3804 patients; screening 120 to 30 min	Endpoint was eligibility, not enrollment or outcome; proprietary; developer-affiliated authors; E4

A. Graph and retrieval metrics must be task-conditioned

Graph accuracy should be broken out by node and edge type, source, extraction method, and evidence grade. A micro-average hides the cases that matter. Frequent, easy relations dominate the number while performance on rare but safety-critical edges, contraindications above all, stays invisible. A useful audit samples from each stratum and asks domain experts to judge the assertion, its direction, its context, its temporality, and its provenance.

Completeness is better framed as task-specific coverage than as some unknowable global quantity. Ask what fraction of evaluation cases have their decisive diagnosis in the top 50 retrieved candidates. Ask what fraction of gold evidence statements the graph schema can even express. Both are answerable; "is the graph complete" is not.

Retrieval evaluation needs sufficiency and parsimony together. Recall at k says whether the gold evidence was retrieved; precision at k says how much of what came back was relevant. For a graph the evidence unit has to be declared first, since an edge, a path, a neighborhood, a community summary, and a source passage are not comparable. Compactness matters too, reported as relevant edges over total retrieved edges alongside the token count. A graph retriever can post high node recall by returning a neighborhood so large the model drowns in it. The BioRAG ablation found no universal winner among retrieval strategies, only trades between faithfulness, context recall, and context precision [85].

For systems that generate SPARQL or Cypher, string match is the wrong measure, because two syntactically different queries can mean the same thing. Report parsing success, execution success, denotation accuracy, and unsafe-query rate instead. A sample of traces should also be adjudicated by hand, since a query can execute cleanly and still ask the wrong question.

B. Reasoning faithfulness is distinct from path plausibility

A retrieved path is not automatically an explanation. Four concepts should stay separate. *Attribution correctness*: the cited edge, path, or document supports the claim. *Retrieval faithfulness*: the system reports what was retrieved and does not invent graph content. *Reasoning faithfulness*: the cited evidence actually influenced the answer rather than being attached afterward. *Clinical validity*: the reasoning and conclusion suit the patient and decision context.

TABLE VII

SEVEN LAYERS AT WHICH A GRAPH-GROUNDED SYSTEM SHOULD BE MEASURED, AND WHY ONE NUMBER IS NOT ENOUGH. A SINGLE ACCURACY FIGURE CANNOT SAY WHERE A SYSTEM SUCCEEDED OR FAILED, BECAUSE THE PIPELINE HAS MANY PLACES TO GO WRONG AND SLACK AT ONE STAGE CAN MASK A DEFICIT AT ANOTHER. A CORRECT ANSWER MAY COME FROM THE MODEL'S PARAMETRIC MEMORY WHILE THE RETRIEVER RETURNED NOTHING USEFUL; A PERFECT GRAPH EARNS NOTHING IF THE MODEL IGNORES WHAT IT IS GIVEN. LAYERS L1 THROUGH L7 RUN IN PIPELINE ORDER, FROM THE GRAPH ITSELF THROUGH CONCEPT LINKING, RETRIEVAL, EVIDENCE USE, GENERATION, OPERATIONAL COST, AND FINALLY WHETHER THE OUTPUT HELPS A HUMAN DO THE WORK. THE FIRST FIVE BELONG IN ANY SYSTEM PAPER. L6 AND L7 BECOME OBLIGATORY ONCE THE CLAIM MOVES FROM ALGORITHMIC PERFORMANCE TO PRACTICAL UTILITY, BECAUSE LATENCY, COST, AND MISPLACED RELIANCE ARE THE PROPERTIES THAT DECIDE WHETHER A SYSTEM SURVIVES CONTACT WITH A WORKFLOW.

Layer	Core question	Recommended measures
L1 Graph and schema	Is the knowledge correct, coherent, current, fit for purpose?	Entity/relation precision and recall by type; ontology conformance; duplicate and contradiction rates; provenance coverage; temporal freshness; task-specific coverage
L2 Normalization and query formation	Right concepts and an executable query?	Entity-linking accuracy; top-k concept recall; ambiguity rate; SPARQL/Cypher exact and execution accuracy; relation-direction accuracy; invalid-query and timeout rates
L3 Retrieval	Sufficient, relevant, economical evidence?	Evidence recall@k, precision@k, MRR/nDCG; answer-bearing path recall; path validity; subgraph compactness; source diversity; latency
L4 Reasoning and evidence use	Did the model use the graph correctly?	Multi-hop/path validity; supported intermediate-claim rate; relation-direction errors; contradiction handling; counterfactual consistency; distractor resistance; sensitivity to removing decisive edges
L5 Generation and attribution	Correct, complete, calibrated, supported?	Task accuracy/F1; claim-level support and contradiction rates; citation precision/recall; completeness; harmful-error severity; Brier/ECE; abstention and risk–coverage
L6 System and operations	Worth the cost and stable under updates?	P50/P95 latency; throughput; token and retrieval volume; cost/query; memory and energy; timeout/failure rate; build/update time; sensitivity to model or graph change
L7 Human and clinical utility	Does it improve the work without unacceptable harm?	Task completion, time, decision change, appropriate reliance, override rate, cognitive load, subgroup effects, adverse-event severity, prospective outcomes

Attribution can be measured directly. Break the response into atomic claims, then label each one as supported, contradicted, or not addressed by the evidence it cites. ALCE established citation recall and precision as separate dimensions [86]. VERICITE showed why that separation matters here: across four models and 500 BioASQ questions, only 27 to 41% of citation pairs were supported at sentence level, and supplying gold documents improved the answers without much improving the citations [11]. A visible identifier is not evidence that the sentence before it follows.

Reasoning faithfulness cannot be measured by inspection. It needs interventions. Remove the decisive edge and see whether the answer changes. Replace it with a controlled counterfactual. Reverse a relation. Add plausible distractor paths. Or keep all the node text and shuffle the topology. That last one is the most diagnostic single test available: if performance survives having the graph structure destroyed, the benefit was coming from the added text, not the relations. If the answer instead follows a false counterfactual edge, the system is graph-sensitive but not robust. Either result tells you more than a fluent chain of thought.

Automated evaluators such as RAGAS and ARES give scalable estimates of context relevance, answer relevance, and faithfulness [87, 88]. They are measurement instruments, not ground truth. A study using one should report its model and version, the prompt, the threshold, the calibration sample, and its agreement with clinician labels. No safety-critical conclusion should rest on a model judging another model.

C. Matched baselines: isolating the value of graph structure

Many published comparisons change several factors at once, giving the graph arm a larger corpus, more context, a stronger generator, or more inference calls than the baseline. Such experiments assess whole products and cannot attribute the gain to the graph. Table VIII sets out the arms and ablations that can, and the constraint that makes them work is holding everything else fixed: the same question set, knowledge cutoff, corpus snapshot, generator, prompt budget, and where feasible the same token and latency budget.

The primary graph-effect estimate should be a paired difference with uncertainty, $\Delta_G = S(\text{graph retrieval}) - S(\text{matched text retrieval})$, for a predeclared metric S. Secondary estimates should normalize by token count, latency, and cost. For open-ended answers, blinded adjudication should compare paired outputs and classify what actually improved: a newly retrieved fact, a better multi-hop connection, improved provenance, or simply more context.

Stress testing belongs alongside the ablations. Useful perturbations include missing decisive edges, poisoned or incorrect edges, retracted evidence, outdated guidelines, contradictory studies, ambiguous acronyms, rare synonyms, unseen relations, and prompt injection inside retrieved text. For patient graphs, add incomplete histories, reordered events, copied-forward notes, and cross-institution terminology drift. Report both average degradation and the worst clinically significant failure, because a system can degrade gracefully on average while failing catastrophically on a narrow class of input.

D. When does graph grounding help, and when does it hurt?

The evidence supports a conditional claim: graphs help when they add nonredundant, sufficiently covered, query-relevant structure that the model can consume within its context and reasoning limits. Table IX summarizes the emerging pattern.

TABLE VIII

The comparison arms and ablations needed to show that the graph, and not something else, produced the improvement. Published comparisons often change several things at once, giving the graph arm a larger corpus, a longer context, or a stronger generator than the baseline, which measures a whole product rather than the graph's contribution. The upper block is a ladder of comparators, each answering one question: the closed-book arm sets the floor the graph must beat, the text-retrieval arms test whether ordinary passage search would have sufficed, and the oracle arm separates a retrieval failure from a generation failure. The lower block removes or corrupts one component at a time. The topology-shuffle ablation is the most diagnostic single test in the table, because a system whose performance survives having its graph structure destroyed was benefiting from the added text, not the relations. Every arm must run against the same question set, snapshot, generator, and context budget, or the comparison reintroduces the confound it was meant to remove.

Arm or intervention	Question it answers	Interpretation
Comparator ladder		
Closed-book model	How much comes from parametric knowledge?	Sets the floor the graph must beat
Long-context input	Is retrieval needed at all at this corpus size?	If not, retrieval is added complexity
BM25 text retrieval	Does lexical matching suffice?	Cheapest baseline that often suffices
Dense or hybrid text retrieval	Does semantic passage retrieval suffice?	The comparator most papers omit
Symbolic graph query	Does the model add anything beyond execution?	Isolates the model's contribution
Graph embedding or GNN	Does language reasoning beat graph-native prediction?	Relevant for link-prediction framings
Graph retrieval	Does topology improve selection or reasoning?	The claim under test
Graph plus text	Is topology complementary to passages?	Usually the strongest configuration
Oracle evidence	What is the ceiling if retrieval succeeds?	Separates retrieval from generation failure
Ablations		
Remove graph context	Total value of graph conditioning	No change means the graph was unused or redundant
Shuffle topology, keep node text	Value of structure beyond added text	Persistence undermines structural claims
Random or degree-matched subgraph	Benefit beyond size and hub priors	Detects gains from generic high-degree concepts
Strip provenance, confidence, time	Whether metadata is actually used	Often reveals it is not
Vary hops and subgraph size	Coverage against dilution	Locates the model-specific optimum
Old versus new snapshot	Value and risk of updating	Measures freshness gain and regression
Oracle versus real entity linking	Contribution of normalization error	Bounds reasoning independent of linking
Disable verifier or arbiter	Contribution of each control	Shows whether safety precedes or follows generation
Drop one source database	Source dependence and redundancy	Exposes hidden reliance on a single source

These findings motivate an adaptive grounding policy rather than unconditional GraphRAG. Before retrieval, the system should estimate entity-linking confidence, graph coverage, knowledge freshness, and whether a deterministic lookup would answer the query. After retrieval, it should estimate evidence sufficiency and contradiction. A low-coverage or conflicting-evidence case should trigger broader text search, a different graph, clarification, or abstention, not simply more graph hops. The objective is not maximum graph use but minimum expected risk under a fixed evidence and compute budget.

E. Stratified evaluation and subgroup validity

A single aggregate score hides several distinct mechanisms by which a KG–LLM pipeline can fail unevenly across populations. The source literature overrepresents some populations and diseases, so graph coverage is not uniform. Terminologies can encode outdated or overly broad categories. Entity linking often performs worse on local, multilingual, or community-specific language. Sparse coverage produces systematic retrieval failure for rare conditions rather than random error. Each of these is a measurable pipeline property, not a downstream social consideration, and each maps onto a layer in Table VII.

Evaluation should therefore stratify the whole pipeline, reporting concept-mapping coverage, retrieval recall, abstention, calibration, error severity, and task outcome across clinically and socially relevant groups, sites, languages, and data-quality strata. Small-group estimates need confidence intervals, and intersectional strata need explicit sample-size reporting rather than silent aggregation. When a demographic attribute is used for clinically justified effect modification, the relation and its supporting evidence should be stated, and a system should not infer biological meaning from a social category by default. The prospective trial-matching study reported subgroup performance alongside a pre-specified alert threshold [68], which is the reporting standard other work should meet; equal aggregate accuracy across groups still does not establish equal benefit when access, referral, or resource availability differ.

TABLE IX

When adding a knowledge graph helps, when it does not, and what the published evidence shows in each case. This is the survey's central empirical claim assembled in one place. The pattern is conditional rather than uniform. Graphs pay off when they hold knowledge the model lacks, when entity linking succeeds, and when the retrieved subgraph is small enough to use, which is why the rows reporting gains involve patient-context structure, typed path reranking, or facts genuinely novel to the model. The lower rows are the counterexamples that a survey should not omit: retrieval producing one- to two-point inconsistent gains, deeper retrieval lowering citation faithfulness, the same graph snippet helping one model and confusing another, and filtering a graph down improving results. Read together they argue for treating graph use as a decision to be made per query rather than a fixed architectural commitment.

Condition	Evidence	Status	Interpretation
Patient context needs structured neighborhood retrieval	KARE reports sizable gains for mortality and readmission on MIMIC-III/IV [44]	Peer rev.	Community structure collects dispersed context; retrospective accuracy is not clinical validation
Diagnosis benefits from entity weighting and path reranking	medIKAL on a Chinese EMR dataset [40]	Peer rev.	Helps when linking and candidate coverage are adequate; dataset- and language-specific
Query is incomplete, benefits from hypothesis-guided expansion	HyKGE on Chinese medical QA [41]	Peer rev.	Parametric knowledge can aid retrieval but can also bias it toward self-confirming paths
Local documents must connect to authoritative sources	MedGraphRAG vs. no-retrieval, text-RAG, GraphRAG [46]	Peer rev.	Graph organization aids source-linked retrieval; sources still need claim-level entailment
Decisive knowledge is outside model training	Little MedQA gain from PrimeKG, near-full recovery on counterfactual/novel facts [32]	Preprint	Marginal value depends on knowledge novelty; public facts may be redundant for strong models; single study, magnitudes indicative
Ontology retrieval has high candidate coverage	Ontology retrieval beats open generation when the true disease is in the candidate set [89]	Peer rev.	Coverage is a gating variable; low coverage should not force a graph-bounded answer
Retrieval adds mostly redundant or noisy context	Small, inconsistent 1–2 point gains; distractors can hurt [10]	Peer rev.	Retrieval success and evidence use are separate bottlenecks
More evidence dilutes attribution	Citation support falls as depth rises [11]	Peer rev.	Tune depth against both coverage and support precision
Short snippets interact with architecture	Snippets help some LLMs, hurt others [12]	Peer rev.	Serialization and interface compliance are part of the treatment
Evidence is heterogeneous and incomplete	CrossDDI: PubMed covered only 40.5% of positives [55]	Peer rev.	Deterministic arbitration aids traceability but cannot recover absent facts
Scientific sources conflict	BioConflict: strong detection, brittle synthesis [90]	Peer rev.	Flattening conflicting edges can be worse than preserving disagreement
Graph has many irrelevant relations	Metapath filtering improved repurposing link prediction [83]	Peer rev.	More graph is not more signal; task-conditioned filtering can help

F. *Reporting negative and null results*

Null graph gains are informative when the comparison is matched. Authors should report per-dataset and per-question-type effects rather than only an aggregate mean, because coverage regimes can cancel out. They should disclose the fraction of cases in which no query entity was linked, no candidate answer was reachable, the decisive evidence was retrieved, retrieved evidence contradicted the final answer, and retrieval flipped a correct closed-book answer to an incorrect one. That last quantity, the harmful answer revision rate, matters most for strong LLMs. Primary reporting should include paired absolute and relative differences with 95% confidence intervals and per-case transition counts (wrong-to-right, right-to-wrong, unchanged). Effects should be stratified by coverage, novelty, hop length, specialty, and model size, with retrieval-conditioned accuracy reported separately when gold evidence is and is not retrieved. Studies should also report supported-claim and contradiction rates, latency, tokens, cost per corrected answer, and severity-weighted clinical error counts. McNemar's test or paired bootstrap intervals suit paired discrete outcomes, while stratified bootstrap or hierarchical models suit clustered questions, and multiple comparisons need a declared primary contrast with correction. Mean scores should come with seeds or repeated API runs, because stochastic decoding and changing hosted models can reorder ranks.

G. *Reproducibility and auditability*

A KG–LLM paper is reproducible only when another group can reconstruct the knowledge boundary and the retrieval behavior, not just rerun a prompt against a similarly named model. Releasing code is insufficient. Reproduction requires the graph snapshot or a deterministic recipe for rebuilding it, the retrieval configuration down to index version and depth, the exact model revision with its access date, and the retrieved identifiers for every test instance. A full artifact checklist, and a per-instance record format, are given in the supplementary material [20].

Reproducibility is better reported as a profile than as a yes or no. A paper that describes its architecture but ships nothing is not in the same position as one that ships code but omits the graph snapshot, and neither is in the same position as one whose main estimates an independent group has reproduced. The most demanding level, and the one that matters for clinical claims, is reproduction on a later time window, a different institution, or a newer snapshot, because that is the only version of the test that survives contact with a changing world. Most current work sits at the lower end: code or prompts are available, the snapshot and traces usually are not. Proprietary APIs do not rule out useful reproducibility, but they do require dated identifiers, preserved raw outputs, repeated runs, and a plan for what happens when the endpoint retires.

Evidence tables should keep reproducibility, peer-review status, and clinical maturity in separate columns, because they measure different things. A reproducible offline benchmark is not clinically mature, and a promising clinician study is often not computationally reproducible.

The evidence should not be pooled into a single "KG–LLM accuracy" number. Tasks, graphs, model generations, baselines, and metrics are too heterogeneous, and many comparisons are confounded. Quantitative synthesis fits only repeated task–dataset–metric combinations with matched interventions; otherwise a structured evidence profile is the honest choice, recording effect direction and uncertainty, baseline strength, coverage, leakage risk, peer-review status, reproducibility, and maturity. The defensible conclusion from the recent literature is bounded. Biomedical graphs can improve retrieval, reasoning, verification, and updating, especially for knowledge the model does not already hold. They do not eliminate hallucination, ensure faithful explanations, or confer clinical validity. Their benefit depends on coverage and quality, the retrieval and serialization interface, the model's ability to use what it is given, and evaluation against a matched alternative.

XI. Applications and the Maturity of Their Evidence

Biomedical KG–LLM systems get grouped under one label even though their uses span very different evidence and risk regimes. A system that proposes a drug–gene association for laboratory follow-up is not the same as one that recommends a diagnosis from an EHR, and neither is the same as a patient-facing assistant. The first produces a research hypothesis, the second can shape professional judgment, and the third can change patient behavior directly. Application claims should therefore be read alongside the intended user, the decision consequence, the evidence maturity, and the degree of human control.

A. Scientific discovery and hypothesis generation

The most natural use of biomedical KGs is to organize heterogeneous evidence across genes, proteins, pathways, phenotypes, diseases, drugs, publications, and trials. LLMs extend this in two directions: they extract candidate entities, relations, and qualifiers from the literature, and they turn graph neighborhoods or paths into hypotheses a researcher can inspect. The graph supplies an explicit search space while the LLM lowers the cost of schema-constrained extraction and natural-language interpretation.

MDKG illustrates the construction direction. Gao et al. [59] combined GPT-assisted pre-annotation and triplet refinement with active learning, named-entity recognition, relation extraction, expert review, and fusion of curated databases. The resource holds more than ten million relations and attaches conditional statements, study-population characteristics, source information, and confidence metadata to literature-derived assertions. Four experts assessed sampled triplets and reported a correctness score of 0.79, and graph-derived representations improved retrospective prediction in UK Biobank experiments. Two lessons generalize beyond the resource. Contextual qualifiers and provenance should be first-class data, not discarded when sentences become triples. And scale does not replace validation: almost one million associations absent from the source databases are candidates, not established facts.

Reasoning agents run the opposite direction, treating the KG as an external computational substrate. Matsumoto et al. [84] introduced ESCARGOT, which combines executable graph-of-thought planning with biomedical graph queries and reported gains over conventional RAG on open-ended questions. Such systems make tool calls and retrieved facts inspectable, but an inspectable trace is not automatically a faithful scientific explanation. A plausible path can reflect confounding, source duplication, extraction error, or a merely associative edge. The appropriate output is a ranked, provenance-linked hypothesis with explicit uncertainty and a proposed validation experiment, not a claim that a mechanism or treatment effect is established. For drug discovery and repurposing the evidentiary chain should be explicit: computational association, then independent corroboration, then expert prioritization, then experimental validation, then preclinical validation, then clinical evaluation. Most studies stop at one of the first three stages, so a survey should report downstream wet-lab or clinical validation separately and should not call an untested graph prediction a discovery.

B. Information access, evidence synthesis, and question answering

Graph-grounded question answering is the most visible application family, covering consumer health information, professional retrieval, literature synthesis, fact checking, guideline navigation, and explanation of relationships. The graph can support entity normalization, query expansion, multi-hop retrieval, relation-aware reranking, and evidence organization before generation. Wu et al. [46] evaluated MedGraphRAG across nine medical QA benchmarks, two fact-checking datasets, and long-form generation, connecting user documents to controlled concepts and authoritative sources through combined top-down and bottom-up retrieval. This is a meaningful advance over flat passage retrieval, but the evidence stays benchmark-level: medical QA scores do not show whether clinicians retrieve the right evidence under time pressure, whether cited passages entail each claim, or whether use changes decisions safely.

The distinction between answer quality and evidence quality is essential. A system can give a correct answer from unsupported internal knowledge. It can give a wrong answer despite retrieving the right evidence. It can give a fluent answer whose citations do not support its claims. Evaluation therefore has to separate four things: retrieval coverage, source quality, claim-to-evidence entailment, and final-answer correctness. Showing sources is not grounding, and a graph path is not an explanation.

Information-access systems suit staged deployment, because their first job can be retrieval rather than recommendation. A low-risk first interface returns an evidence packet: normalized concepts, dated sources, supporting and conflicting passages, and the graph paths between them. No treatment recommendation, no synthesis. Generation gets added later, once retrieval quality and citation faithfulness have been validated. Staging also makes failure easier to localize, since an error in the packet is visibly a retrieval error rather than a reasoning one.

C. EHR-based diagnosis, prognosis, and decision support

EHR applications promise context-specific reasoning and face the hardest combination of problems: poor data quality, temporal complexity, privacy, and workflow risk. Two strategies dominate. One turns a patient record into graph-linked concepts and retrieves disease-centered paths. The other retrieves patient-relevant graph communities and uses them as context for prediction. DR.KNOWS takes the first route, mapping concepts in clinical notes to UMLS and ranking knowledge trajectories toward a diagnosis [39]. medIKAL adds typed entity weighting and path reranking, and merges an initial model diagnosis with graph search [40]. KARE takes the second route, using hierarchical communities to enrich patient representations for mortality and readmission prediction [44].

These studies show that graph structure can add useful inductive bias and patient-specific context. They do not establish clinical effectiveness. MIMIC experiments are retrospective and largely single-center, and a new EHR benchmark is not an external validation just because its records came from real practice. The graph also opens new leakage routes. If an answer label, a diagnosis code, a future event, or an edge closely derived from one sits in the snapshot, good performance may reflect access to the target rather than clinical reasoning. A temporal split has to lock all of it: the record, the literature, the terminology releases, and the graph snapshot.

Clinical meaning also lives in the qualifiers. Consider what the triples *(patient, has_condition, disease)* and *(drug, treats, disease)* throw away. The source may have said "family history of," or "ruled out," or "previously treated," or "contraindicated in pregnancy," or "supported only in a biomarker-defined subgroup." Each of those changes the clinical answer. A patient graph therefore needs to model assertion status, who the statement is about, the time interval, the source, and the confidence. A global graph needs to keep population and evidence qualifiers. Without them, a retrieval can be technically perfect and clinically wrong.

D. Cohort identification and clinical-trial matching

Trial matching fits neuro-symbolic architectures well, because eligibility combines free text with exact constraints on diagnoses, biomarkers, prior therapies, laboratory values, performance status, geography, and time. LLMs can extract and normalize evidence from records and protocols, graphs can link synonyms, therapies, variants, diseases, sites, and trials, and deterministic components can evaluate numeric, Boolean, and temporal rules.

The strongest clinical-stage evidence in the recent corpus is the prospective evaluation by Loaiza-Bonilla et al. [68]. Their oncology platform combined domain-specific LLM agents, an oncology KG, deterministic eligibility reasoning, a recommendation engine, and oncologist review. Across 3,804 prospectively screened patients it processed 23,912 candidate patient–trial pairs and reached an F1 of 0.82, against 0.47 for a GPT-4 zero-shot baseline and 0.67 for a GPT-4 chain-of-thought baseline. Median screening time fell from about 120 minutes for manual review to about 30 minutes including clinical verification, and the study reported calibration, ablations, and subgroup analyses. This prospective evaluation moves beyond offline benchmarking, while its limits identify the next evidentiary step. The unit of evaluation was eligibility classification, not enrollment, participation, survival, or patient-reported outcome. Cases needing clinician judgment were excluded from the binary F1, and rare biomarker combinations, temporal criteria, and incomplete extraction remained error sources. The platform was proprietary and several authors were affiliated with its developer. Independent multisite replication, comparison with the existing screening workflow, prospective error surveillance, and measurement of actual enrollment and equity effects are still needed, so the study supports clinician-supervised operational feasibility rather than patient benefit.

E. Patient-facing systems

Patient-facing advice is the highest-risk application, because users may act without professional mediation, may disclose sensitive information, and may be unable to judge whether an answer is well grounded. The WHO guidance on large multimodal models in health identifies patient-guided use as a major application while warning about inaccurate, biased, incomplete, and overly authoritative output and automation bias [91]. A KG can improve source access and terminology control, but it does not resolve these risks, and it can make advice look more authoritative than the evidence warrants. Deployment should be bounded by intended use: educational explanations, service navigation, and preparing questions for a clinician differ from diagnosis, triage, medication adjustment, or treatment selection. Higher-risk functions need conservative scope limits, explicit emergency escalation, calibrated abstention, age- and literacy-appropriate communication, claim-level sources, privacy-preserving interaction, and an accessible route to human care. Evidence from clinician-facing or exam-style QA does not transfer here without direct usability and safety evaluation in the intended patient population.

F. Cross-application synthesis

Table X summarizes the graph's contribution, the appropriate near-term output, and the evidence required before stronger claims for each application. Graphs help most when a task genuinely depends on typed relations, normalization, multi-hop structure, or computable constraints. They may add little when a current, authoritative passage answers the question

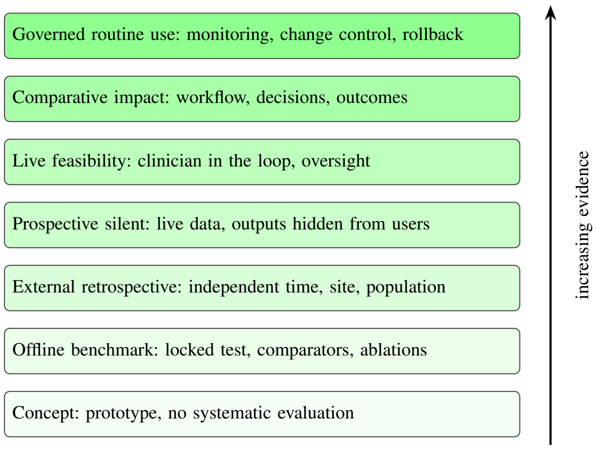

Fig. 8. **Evidence maturity: how far a system's evaluation is from real clinical use.** The ladder grades the evidence, not the ambition, so a paper can target a clinical task and still sit near the bottom. E0 is a working prototype with no systematic evaluation. E1 is an offline benchmark with a locked test set and ablations, which supports a claim about that dataset and none about generalization. E2 repeats the evaluation on an independent time period, site, or population, testing whether the result travels. E3 runs the system on live data in the real workflow with its outputs hidden from users, which surfaces latency, missing documents, and terminology drift that a benchmark cannot. E4 shows the output to clinicians under predefined oversight, testing whether it can be used safely. E5 asks the harder question of whether using it changes decisions or outcomes. E6 is routine use sustained by monitoring, change control, and revalidation, which is an operating state rather than a certificate. The distribution of the reviewed literature is the point of the figure: most systems sit at E1, the single prospective clinical study reaches E4 for one screening workflow, and nothing in the corpus reaches E5.

directly, and they can hurt when coverage is poor, mappings are wrong, or retrieved subgraphs fill the context window without adding decisive evidence.The conditions under which each of these holds, and the comparator ladder that separates them, are given in Table IX and Table VIII.

G. How mature is the evidence?

Words like "clinical," "real-world," and "validated" are used loosely enough across this literature that application claims are hard to compare at all. The ladder in Fig. 8 grades the evaluation rather than the ambition, so a study can address a clinical task and still sit near the bottom of it. E0 is a concept with no systematic evaluation. E1 is an offline benchmark with a locked test set, comparators, and ablations, which says something about the dataset and nothing about generalization. E2 adds external retrospective validation on an independent period, site, or population. E3 is a prospective silent deployment on live data with outputs hidden from users. E4 is clinician-in-the-loop feasibility with predefined oversight. E5 is a comparative study measuring workflow, decisions, safety, or outcomes. E6 is governed routine use, sustained by monitoring, change control, and revalidation.

Each step tests a property that the level below left unexamined. Progression therefore requires different evidence, not simply more of the same. A system that excels on a locked benchmark can fail in silent deployment because local terminology mapping, document availability, latency, or evidence freshness differ from the benchmark environment. A system that clears silent deployment can still increase workload or invite misplaced reliance once clinicians see its output, because accuracy and usability are different properties. Treating a step as a formality is the common failure. Freezing the configuration, running matched comparators, and using temporal rather than random splits are what make the move from benchmark to external validation meaningful. Detailed gate criteria are in the supplementary material [20].

Placing the reviewed work on this ladder is sobering. KARE, medIKAL, DR.KNOWS, MedGraphRAG, and most of their peers sit at E1. The oncology trial-matching study reaches E4 for one defined screening workflow, because clinicians reviewed live recommendations, and it does not demonstrate patient outcomes [68]. Nothing in the corpus reaches E5. The field therefore has substantial algorithmic promise and very little clinical-effectiveness evidence, and abstracts should say so.

The top of the ladder is not a certificate either. Governed routine use is an operating state, held only while monitoring, change control, and revalidation continue, so a system that stops meeting those conditions has left the level whatever its original evaluation showed.

H. What a KG–LLM study should report

No single reporting guideline covers an adaptive KG–LLM system, and the established instruments are named in Section XII. Because those guidelines predate graph-grounded generation, they omit several KG–LLM-specific items. A study should additionally report the graph schema and snapshot identifier, the retrieval corpus and configuration, the serialization used in the prompt, the tool policy, the evidence-verification path, the ablations that isolate the graph effect, terminology releases and mapping versions, the leakage analysis, graph-coverage strata over the evaluation set, and the exact model identifier with its access date. Together these define the intervention that was evaluated, which is why the configuration tuple of Section XII is a reporting requirement rather than an operational convenience.

Measured performance also belongs to the human-machine team rather than the model alone, which makes the interface an experimental variable. Useful transparency exposes the basis for verification, not a narrative of how the model supposedly reasoned: the patient facts used, their dates and assertion status, the concepts they mapped to, the retrieved sources with version and date, any material conflict, and the confidence or abstention status. Oversight has to be designed and measured rather than asserted, since adding an accept button transfers liability without establishing review. If a user cannot evaluate the basis of a recommendation in the time available, the human-in-the-loop control is nominal.

XII. Failure Modes, Security, and Operational Control

The preceding sections examined what KG–LLM systems can do. This section addresses their failure modes and the engineering controls that contain the resulting damage. A deployed KG–LLM system is a versioned configuration whose knowledge substrate changes over time, making configuration

TABLE X

What the graph contributes in each application area, what such a system can defensibly output today, and what evidence a stronger claim would require. The applications differ less in technique than in what a mistake costs. A drug–gene association proposed for laboratory follow-up is a hypothesis a researcher will test; a diagnosis suggested from a patient record can shape professional judgment; advice given directly to a patient may be acted on with no clinician in between. The middle column is therefore deliberately conservative, naming the output that current evidence supports rather than the one the technique could eventually produce, and the right column names what is missing. The pattern across rows is that graphs help most where a task genuinely turns on typed relations, normalization, or computable constraints, and least where a single current authoritative passage would answer the question directly.

Application	Coupling	Graph contribution	Appropriate near-term output	Evidence before stronger claims
Knowledge construction	C1–C2, C5 for promotion	Normalization, structure, qualifiers, provenance	Quarantined candidate assertions for curation	Relation-level validation, inter-rater reliability, source-stratified error analysis
Discovery and repurposing	C2–C4	Multi-hop hypothesis space and evidence linkage	Ranked hypothesis with provenance and validation plan	Independent replication, wet-lab or preclinical confirmation, prospective utility
Literature and guideline access	C1–C2	Concept expansion, retrieval, conflict organization	Evidence packet with dated supporting and conflicting sources	Retrieval recall, citation entailment, temporal freshness, clinician task evaluation
Diagnosis and prognosis	C1–C2 if audited, C3 if throughput dominates	Patient-to-concept mapping, disease paths, community context	Clinician-facing differential or risk estimate with abstention	Temporal/external validation, calibration, subgroup performance, silent then prospective deployment
Trial matching	C5, deterministic verifier	Ontology normalization and deterministic eligibility constraints	Candidate trials plus criterion-level evidence for expert review	Prospective multisite accuracy, operational impact, enrollment, equity analysis
Patient communication	C1, bounded scope	Terminology control and authoritative source retrieval	Bounded education and navigation	Patient safety, comprehension, reliance, accessibility, escalation, outcomes

management, lineage, and adversarial robustness first-order technical concerns rather than deployment paperwork.

A. Semantic mapping as a model-risk source

Interoperability is often reduced to "the system uses FHIR" [92] or "it maps to OMOP" [93]. Neither claim tells you whether meaning survived the mapping, and meaning is what the graph is for. Mappings from local terms to SNOMED CT, ICD, RxNorm, LOINC, UMLS, or OMOP concepts can be one-to-many, approximate, version-dependent, or context-sensitive. Every graph node derived from a patient record should therefore retain the source code and text, the mapped concept, the mapping method and confidence, the terminology version, and the effective time. Unmapped and ambiguous concepts must stay visible, because silently dropping them produces false reassurance, and unit conversion, negation, family history, uncertainty, and temporal intervals all need explicit handling.

The testable consequence is that conformance is not correctness. Interoperability testing should measure clinical meaning through cross-site phenotype agreement, concept-coverage and ambiguity rates, unit and temporal consistency, preservation of source qualifiers, round-trip tests, and performance stratified by mapped versus unmapped concepts. A schema-valid message can still be clinically wrong, and a conformant analytic table can still reflect a lossy extraction pipeline.

B. The deployed system is a versioned configuration

The intervention is not "the LLM." It is the configuration

$$C = (M,\ P,\ G,\ T,\ R,\ V,\ I), \tag{3}$$

where M is the model and version, P the prompt and tool policy, G the KG snapshot and schema, T the terminology and ontology releases, R the retrieval and reranking policy, V the verification and abstention layer, and I the interface and workflow integration. A change to any element produces a new configuration unless equivalence is demonstrated.

This formulation closes a reproducibility gap that version-controlling the model alone leaves open. Updating a drug database, swapping an embedding model, adding an ontology synonym, altering a hop limit, or changing where a result appears in the interface can all change behavior while M stays fixed. Release documentation should therefore carry immutable component identifiers, graph build manifests, source checksums, evaluation results, and rollback targets, and every logged output should name the configuration that produced it. Without this, a regression cannot be attributed and a rollback cannot be verified.

C. Provenance, contradiction, and time as representation problems

Biomedical facts are rarely timeless binary relations. An edge may come from a randomized trial, an observational association, a preclinical experiment, a database inference, a text-extraction model, or an LLM proposal, and its validity may be population-specific, dose-specific, disputed, superseded, or retracted. A graph intended to ground clinical claims should represent at least source identity, publication or record date, extraction method, evidence type, population and experimental qualifiers, confidence, review status, and valid time. MDKG shows what this buys, attaching conditional statements, baseline population attributes, source information, and confidence to relations [59].

Two representation choices follow. First, contradictions should be preserved rather than collapsed into a single consensus edge, so that retrieval can return supporting, opposing, and insufficient evidence and generation can report the disagreement together with the date through which evidence was searched. Second, retraction has to propagate. Removing one document from an index is not enough, because dependent edges, community summaries, cached embeddings, generated rationales, and downstream derived assertions all need to be identified and invalidated or rebuilt. This is a lineage problem, and solving it is what lets an operator answer, after an incident, which outputs used the affected evidence and which configuration should replace them.

D. Hazard analysis across the lifecycle

Failure-mode analysis becomes actionable only when each hazard links to a control and to objective verification evidence. Table XI instantiates this for the lifecycle of Fig. 3. Severity is task-dependent: a missing literature citation and a missed drug contraindication do not warrant the same control budget, so the table should be re-instantiated per intended use rather than applied uniformly.

Section VII explains the mechanism of circular self-validation, the hazard in the closed-loop update row of Table XI, and its four mandatory controls. That row lists the verification evidence a reviewer or auditor should request: a lineage query, a simulated circularity test, and a reversible update drill.

E. Privacy as a graph-structural problem

Patient-specific graphs can be more disclosive than the records they derive from. Linking rare diagnoses, dates, locations, family relations, genomic variants, and social factors eases re-identification even after direct identifiers are removed, and the attack surface extends past the graph itself to embeddings, vector indexes, cached prompts, retrieved subgraphs, logs, and human feedback, all of which belong in the data inventory. Legal obligations vary by jurisdiction and deployment, with HIPAA [94] and the GDPR [95] setting the applicable duties in the United States and the European Union. The engineering point is that neither is satisfied by a compliance label attached to a system whose graph leaks structure.

The architectural response follows from that observation. Keep the global biomedical graph separate from patient graphs, bind access by purpose and role to the smallest subgraph that answers the question, and authorize at query time rather than trusting a downstream filter. Everything derived from a patient graph inherits its sensitivity, so embeddings, cached context, and logs need the same retention limits and egress controls as the records themselves. The full architectural checklist is in the supplementary material [20]. Federated deployment reduces central data movement without guaranteeing privacy, because model updates, gradients, graph statistics, and query patterns can each leak, so a privacy-enhancing technology should be evaluated against a stated threat model rather than adopted as a label.

F. Security and adversarial robustness

Graph grounding changes the attack surface rather than removing it. Yang et al. [96] showed that a single maliciously generated abstract could substantially shift drug–disease rankings in a literature-derived medical KG. Alber et al. [81] showed that poisoning a small fraction of medical training content raised harmful output while leaving standard benchmark performance largely unchanged. PoisonedRAG showed that a few injected documents can steer RAG answers [97]. Together these results refute the assumption that retrieval from an external corpus is intrinsically safer than parametric memory, and they explain why benchmark accuracy is not a safety measurement.

A KG–LLM threat model spans four planes. The *knowledge plane* covers poisoned publications, compromised databases, false triples, ontology manipulation, entity hijacking, malicious summaries, and stale evidence. The *control plane* covers direct and indirect prompt injection, malicious instructions embedded in retrieved text, unsafe query translation, tool-policy bypass, and unauthorized graph updates. The *data plane* covers cross-patient retrieval, inference from embeddings, log leakage, overbroad traversal, and insecure backups. The *action plane* covers unvalidated output passed to orders, messages, scheduling, code execution, or other agents, along with excessive permissions and resource-exhaustion attacks.

Controls should address each of these four planes. Ingestion uses source allowlists, cryptographic manifests, provenance thresholds, anomaly detection, and quarantine. Retrieved content is treated as untrusted data and kept separate from system instructions, so that a document cannot redefine tool permissions. Generated graph queries are parsed against an allowlisted grammar, executed read-only with least privilege, bounded by time and result size, and checked for protected-data scope. Agent tools carry explicit capability boundaries, confirmation for consequential actions, egress restrictions, and complete audit logs. High-risk claims are verified against an independently curated source set rather than the graph that generated them. Security evaluation should report attack success rate and residual harm, not accuracy after perturbation, and safe failure means abstention or a bounded evidence response rather than a silent fallback to ungrounded model knowledge. The NIST Generative AI Profile offers a cross-sector structure for organizing this work across the lifecycle [98].

G. Monitoring a system whose knowledge mutates

A single accuracy measure is too slow and too coarse to catch KG–LLM degradation, because the causes sit at different stages and a drop at one stage can be masked by slack at

TABLE XI

HOW A GRAPH-GROUNDED SYSTEM FAILS AT EACH STAGE, AND WHAT EVIDENCE WOULD SHOW THE CONTROL IS WORKING. EACH HAZARD IS PAIRED WITH A REQUIRED CONTROL AND MEASURABLE VERIFICATION EVIDENCE. THE STAGES ARE ORDERED BY WHEN THEY OCCUR, AND THE FAILURES PROPAGATE DOWNWARD: A BAD EDGE ADMITTED AT INGESTION BECOMES A RETRIEVAL CANDIDATE, A RETRIEVAL GAP BECOMES AN OMITTED CONTRAINDICATION, AND AN UNREVIEWED WRITE-BACK BECOMES TOMORROW'S APPARENTLY INDEPENDENT EVIDENCE. SEVERITY IS TASK-DEPENDENT AND THE TABLE SHOULD BE RE-INSTANTIATED PER INTENDED USE, BECAUSE A MISSING LITERATURE CITATION AND A MISSED DRUG INTERACTION DO NOT WARRANT THE SAME CONTROL BUDGET.

Stage and hazard	Consequence	Required controls	Verification evidence
Source ingestion: outdated, retracted, or malicious source	False or obsolete assertions	Source allowlist and quality tiers; retraction feed; checksums; independent-source rule; quarantine	Source audit; retraction drill; % edges with complete provenance; rebuild test
Construction: entity or relation error	Wrong traversal and downstream output	Schema validation; terminology constraints; calibrated extraction; dual-model or human review for high-risk edges	Relation precision/recall by type; calibration; inter-rater agreement; sampled edge audit
Patient mapping: negation, experiencer, date, unit error	Wrong patient state and unsafe personalization	Assertion and temporal model; unit validation; source-text trace; ambiguity retention	Mapping coverage; temporality/negation test set; cross-site audit; expert adjudication
Retrieval: incomplete or irrelevant subgraph	Omitted contraindication or misleading context	Hybrid retrieval; minimum sufficiency; conflict retrieval; context budget; abstention	Recall@k; path validity; contraindication-recall stress test; oracle comparison
Generation: unsupported inference	Hallucinated diagnosis or recommendation	Claim decomposition; graph and text verification; constrained output; severity-aware abstention	Unsupported-claim rate; citation entailment; counterfactual and distractor tests
Human use: automation bias or alert fatigue	Inappropriate acceptance or workarounds	Role-based actions; calibrated uncertainty; escalation paths	Override and verification rates; time, load, and missed-error analysis
Closed-loop update: proposal re-enters corpus	Circular self-validation and error amplification	Candidate-edge quarantine; independent corroboration; immutable commits; no self-citation	Lineage query; simulated circularity test; reversible update drill
Operations: drift, outage, unapproved change	Silent degradation or uncontrolled behavior	Monitoring; configuration registry; safe fallback; rollback	Drift dashboard; failover exercise; configuration attestation; rollback recovery time

another. Monitoring has to be stage-local. The signals that matter divide along the same lines as the evaluation layers: whether inputs are still mapping to concepts, whether the graph is still fresh and internally consistent, whether retrieval is still returning evidence, whether generation is still supported by what came back, and whether users are still behaving as though they trust it appropriately. A worked signal list is in the supplementary material [20].

Each threshold should be set in advance and mapped to a specific response: investigation, rollback, suspension, or a restricted mode. Sentinel cases should be rerun after every material change to C, so that a series of individually harmless updates cannot shift the system without detection. For a dynamic evidence graph, a high-risk system should retrieve from approved immutable snapshots and promote a new snapshot only after regression testing, because retrieving from a live graph means the system's behavior changes without a release.

Incident response needs traceability in both directions. From a harmful output, an investigator has to reach the patient inputs, mappings, retrieved edges, model, policy, and user actions that produced it. From a corrupted source, they have to reach every release and output that depended on it. Systems that support only the first direction can explain a single failure but cannot identify the full set of affected outputs.

H. Deployment obligations beyond engineering

Engineering controls do not exhaust what a deployment requires. A KG–LLM system used in care falls under medical-device regulation when its intended purpose makes it a device, and under data-protection, professional, and institutional duties regardless. The relevant instruments include the WHO guidance on large multimodal models in health [91], FDA guidance on clinical decision support software and on life-cycle management for AI-enabled device software together with the predetermined change control framework [99, 100, 101], the IMDRF and FDA good machine learning practice principles [102], and the EU AI Act [103]. Reporting guidance is similarly established, spanning TRIPOD-LLM and TRIPOD+AI for development and evaluation, PROBAST+AI for risk of bias, STARD-AI for diagnostic accuracy, DECIDE-AI for early live evaluation, SPIRIT-AI and CONSORT-AI for trials, and FUTURE-AI for deployment [104, 105, 106, 107, 108, 109, 110, 111].

Two implications matter for an engineering reader. The configuration C is what these frameworks govern, so a change to the graph snapshot or the retrieval policy can constitute a regulated change even when the model weights are untouched. And responsibility is distributed across source licensors, graph curators, model providers, system integrators, health systems, and clinical users, so no single actor can certify the whole chain. Detailed accountability allocation and jurisdiction-specific classification lie outside the scope of this survey; the clinical-informatics reviews compared in Section I-A treat them directly [15, 16].

XIII. Future Directions

The next stage should establish when a graph is necessary, whether the whole system is safe, and how its behavior stays controlled as the underlying knowledge changes. Eight priorities follow from the evidence reviewed here, ordered by how soon they are achievable rather than by importance.

Matched comparisons. Run the comparator ladder and the ablations of Table VIII as a matter of routine rather than as a robustness appendix. The field needs a decision rule that predicts which query types benefit from a graph. Current evidence does not yet provide one that practitioners can apply.

Temporal and multisite validation. Freeze all evidence at the prediction time and test on later patients, later literature, and a second institution. Such evaluations remain rare. ChronoMedKG [13], at the time of writing a preprint, shows why they matter by reporting substantial degradation on temporal questions.

Claim-level evidence evaluation. Annotate whether each generated claim is supported, contradicted, outdated, or simply absent from what was retrieved, and whether the cited source is the original. VERICITE showed how far current systems are from this [11], and until it is routine, citation display will keep passing for grounding.

Update metrics for graph-maintaining agents. Agents that write to the graph are evaluated on question-answering accuracy, which measures the wrong thing entirely. What is needed is candidate-edge precision by relation type, conflict-resolution accuracy when sources disagree, time from published correction to graph correction, the propagation radius of a bad edge, and demonstrated rollback. None of these appears in the current literature.

Security benchmarks for biomedical graph retrieval. The poisoning results [96, 81, 97] establish the threat but do not evaluate defenses or residual risk. The field needs healthcare-specific attack suites covering malicious papers, false triples, synonym attacks, indirect prompt injection, and cross-patient leakage, reporting residual risk after defense rather than accuracy under attack.

Agents on the graphs that can support them. The literature-scale and multi-omics graphs carry resolvable provenance and temporal metadata, and no agentic system uses them. An agent over a scholarly-evidence graph could trace a claim from guideline to trial to patent. This workflow requires capabilities that already exist, but no current system combines them.

Multimodal graphs with modality-level attribution. Attaching sequences, structures, images, and signals to graph nodes is now routine [71, 50, 8], and EEG-MedRAG extends retrieval to physiological time series [72]. What no system reports is which modality carried a given conclusion. A multimodal system should be able to state that a prediction rested on the structure rather than the text, and should be ablated one modality at a time, because a fused embedding hides exactly the attribution a clinician needs. Aligned representations also inherit the coverage and provenance problems of every contributing source at once, so the data card in the supplementary material [20] has to be filled in per modality rather than per graph.

Trustworthy deployment as an engineering target. Security is one property among several. Privacy, human oversight, change control under regulatory constraint, and post-deployment monitoring are treated in Section XII and the supplementary material as requirements, but almost no reviewed system reports evidence against them. The gap between the controls that are specifiable and the controls that are demonstrated is now the binding constraint on clinical deployment, and closing it is an evaluation problem rather than a modelling one.

Beyond these, the field would benefit more from shared evidence infrastructure than from another leaderboard: a versioned record linking each system to its model, graph, snapshots, comparator, metrics, failure modes, and released artifacts, with benchmark cases that carry conflicting evidence, time stamps, and security perturbations. That would replace the question of whether a graph improved a model on a benchmark with the more useful one: for which users, tasks, and evidence conditions does a given configuration pay for itself, and what keeps it paying.

XIV. Conclusion

Biomedical knowledge graphs and language models are complementary, yet the graph is neither an automatic truth source nor an explanation certificate. The recent literature shows real advances in graph retrieval, community and path reasoning, patient-context prediction, graph-aware representations, model-assisted curation, and verification-first architectures. It also shows that the gains are conditional. Missing or stale knowledge, entity-linking error, poor subgraph selection, lossy rendering, weak evidence use, contradiction, and ungoverned updates can erase the benefit or leave a system less reliable than a simpler alternative.

Progress depends on changing the unit of evaluation from the generated answer to the complete, versioned lifecycle. Studies should establish the graph's causal contribution with matched baselines and ablations, evaluate quality, retrieval, reasoning, generation, operations, and human use separately, preserve provenance and time, and report evidence maturity without equating retrospective benchmarks with clinical validation. Closed-loop systems need quarantined updates, independent verification, reversible versions, and explicit responsibility.

The field should now focus on whether the resulting systems can be inspected, challenged, updated, reproduced, and safely governed. Meeting those criteria would move graph-augmented systems beyond demonstrations and toward trustworthy biomedical AI engineering.

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